

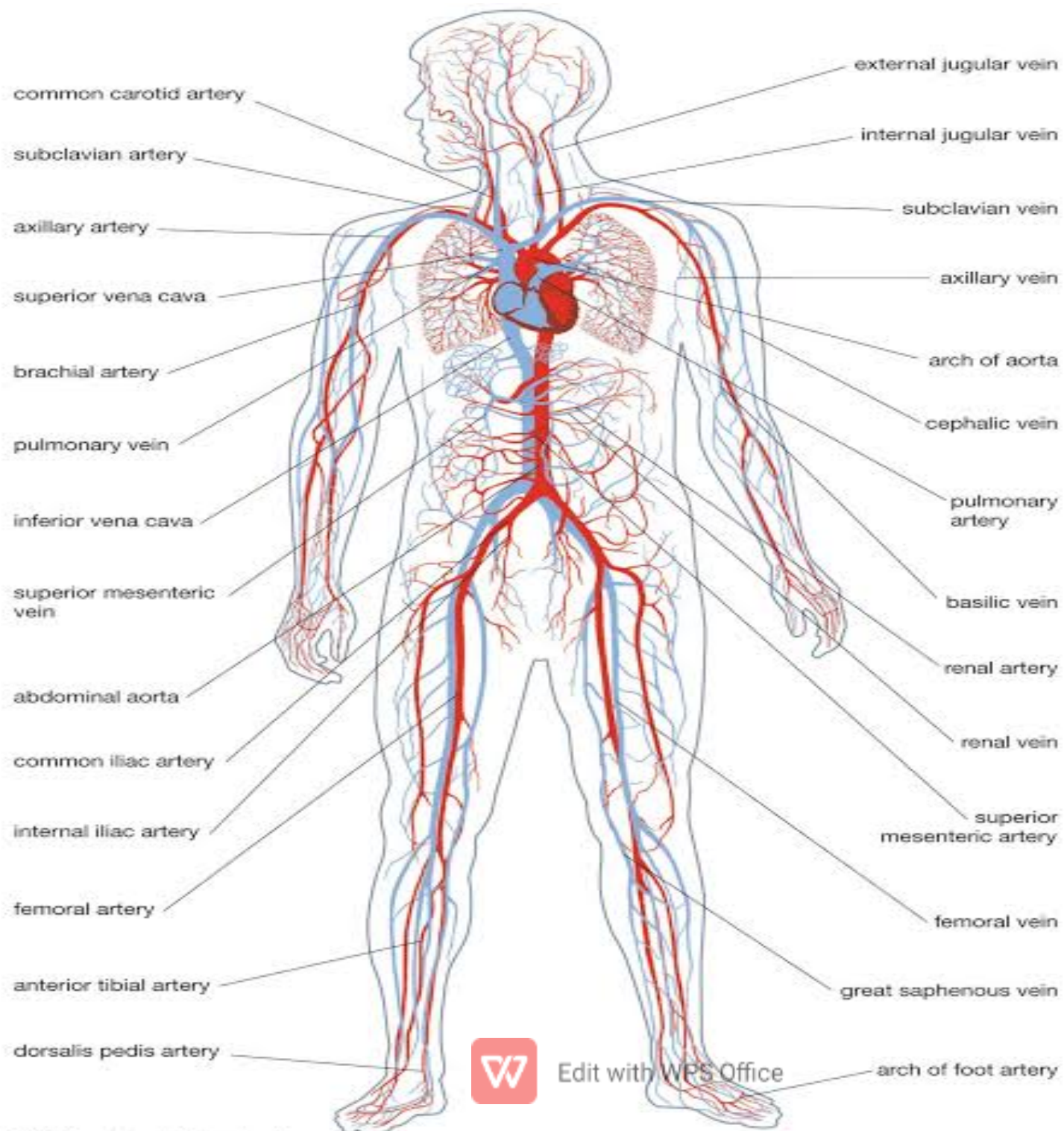
CIRCULATORY ANATOMY



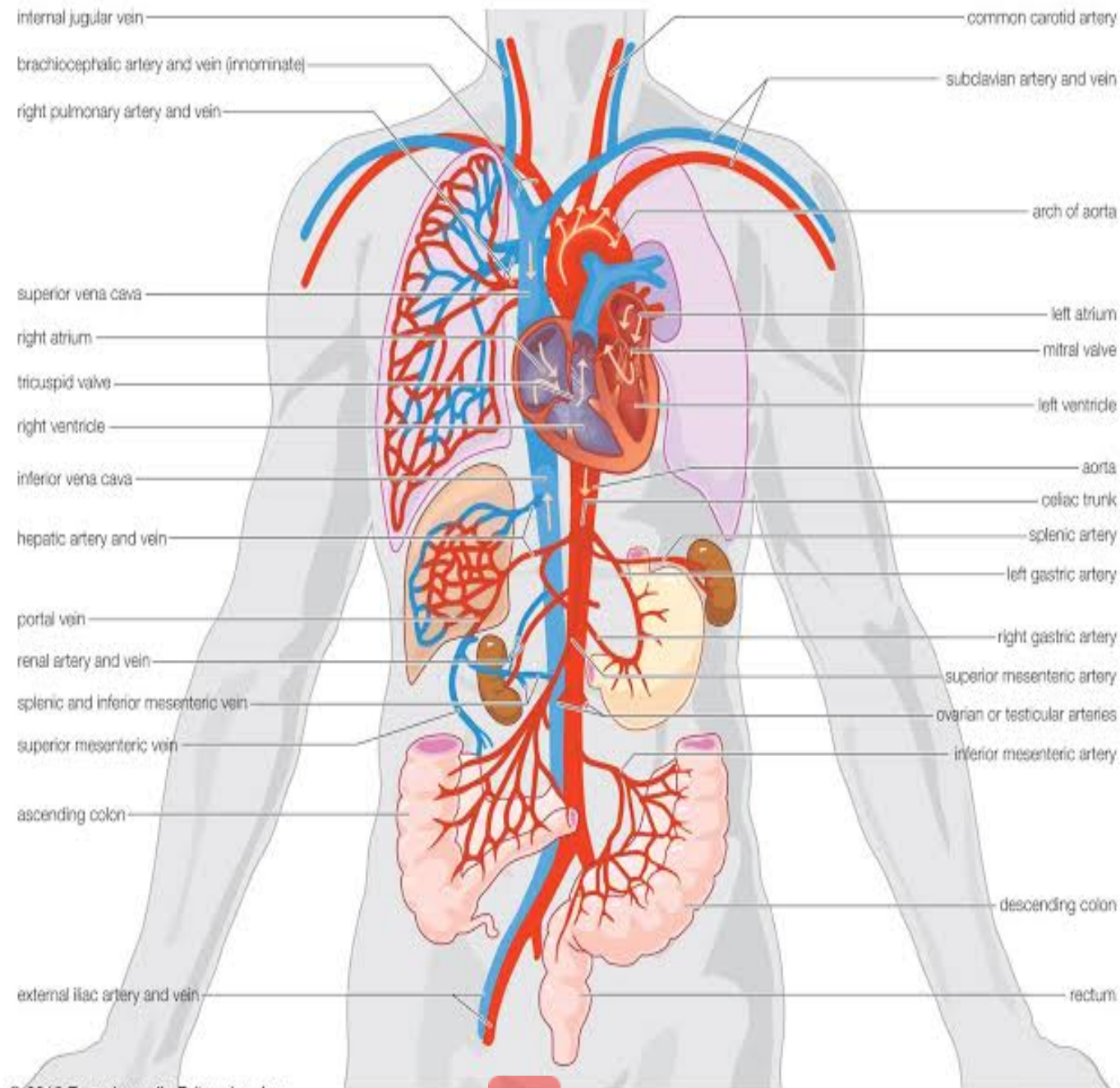
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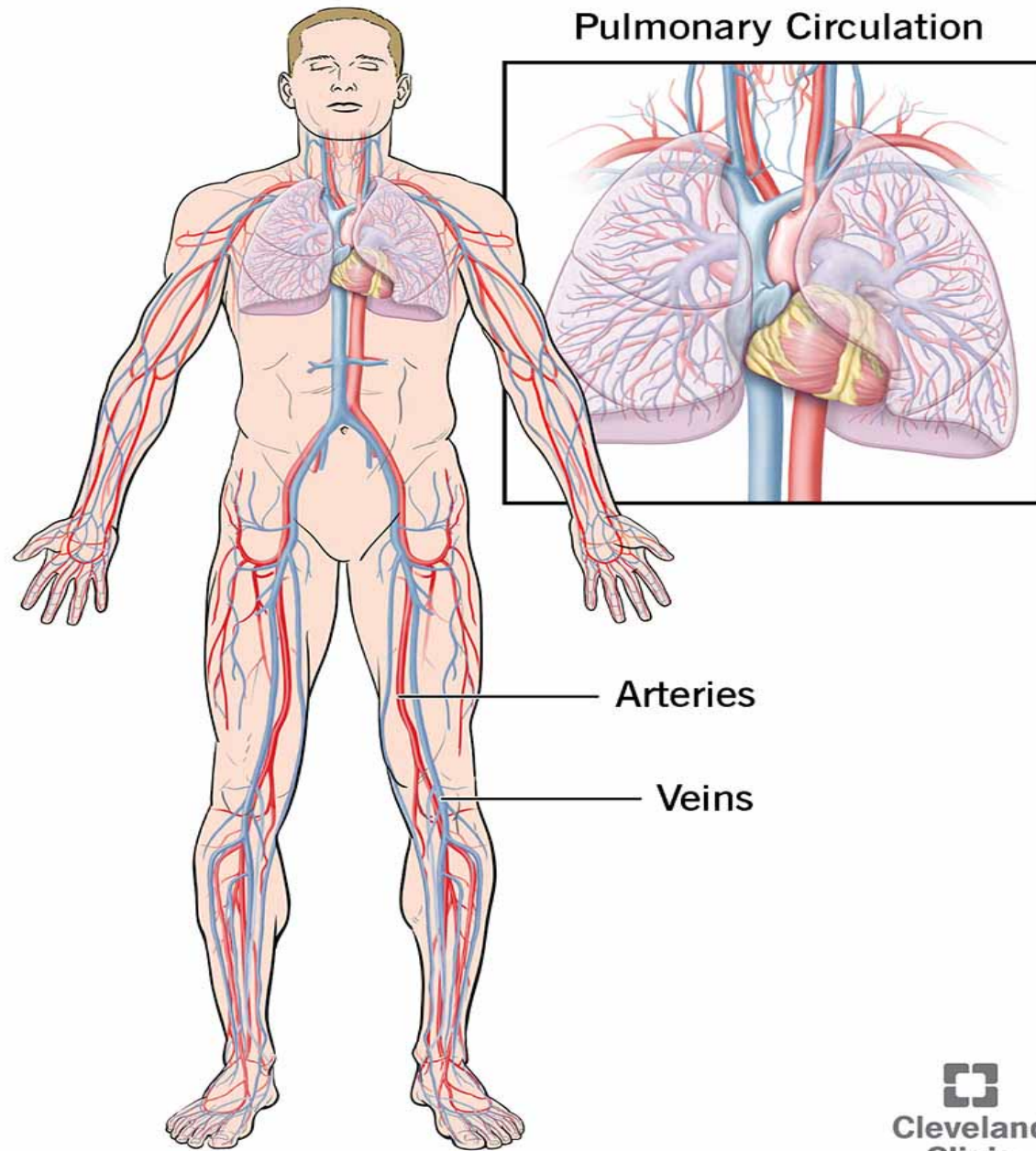
- ✘ Circulatory system is the system which contains the structures that pumps the blood to whole part of the body.





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Systemic Circulation

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BLOOD

- ✧ The blood is a fluid connective tissue that transports nutrients, hormones, enzymes and other metabolic waste products in appropriate place.
- ✧ The blood acts as the medium in circulatory system.
- ✧ The central organ which is responsible for the pumping of blood into the blood vessel is known as heart.



PARTS OF THE CIRCULATORY SYSTEM

- ☒ Heart
- ☒ Arteries
- ☒ Veins
- ☒ Capillaries



COMPONENTS OF BLOOD

- ✧ The blood has 4 components namely,
- ✧ 1. Plasma
- ✧ 2. Red blood cells or erythrocytes.
- ✧ 3. White blood cells or leucocytes
- ✧ 4. Platelets.
- ✧ In this the plasma is fluid and liquid part of blood. All other are the blood cells.



FUNCTIONS OF BLOOD

- ✕ 1. Respiratory function
- ✕ 2. Nutritive function
- ✕ 3. Excretory function
- ✕ 4. Transport function
- ✕ 5. Maintain water balance
- ✕ 6. Storage function
- ✕ 7. Defense function



PERICARDIUM

- ⊠ Pericardium is a fibro-serous sac which encloses heart and it's vessels.
- ⊠ This is situated in the middle mediastinum seated over the dome of diaphragm.
- ⊠ This is the conical shaped structure.
- ⊠ The apex is facing upwards and the base is seen inferiorly.



LAYERS OF PERICARDIUM:

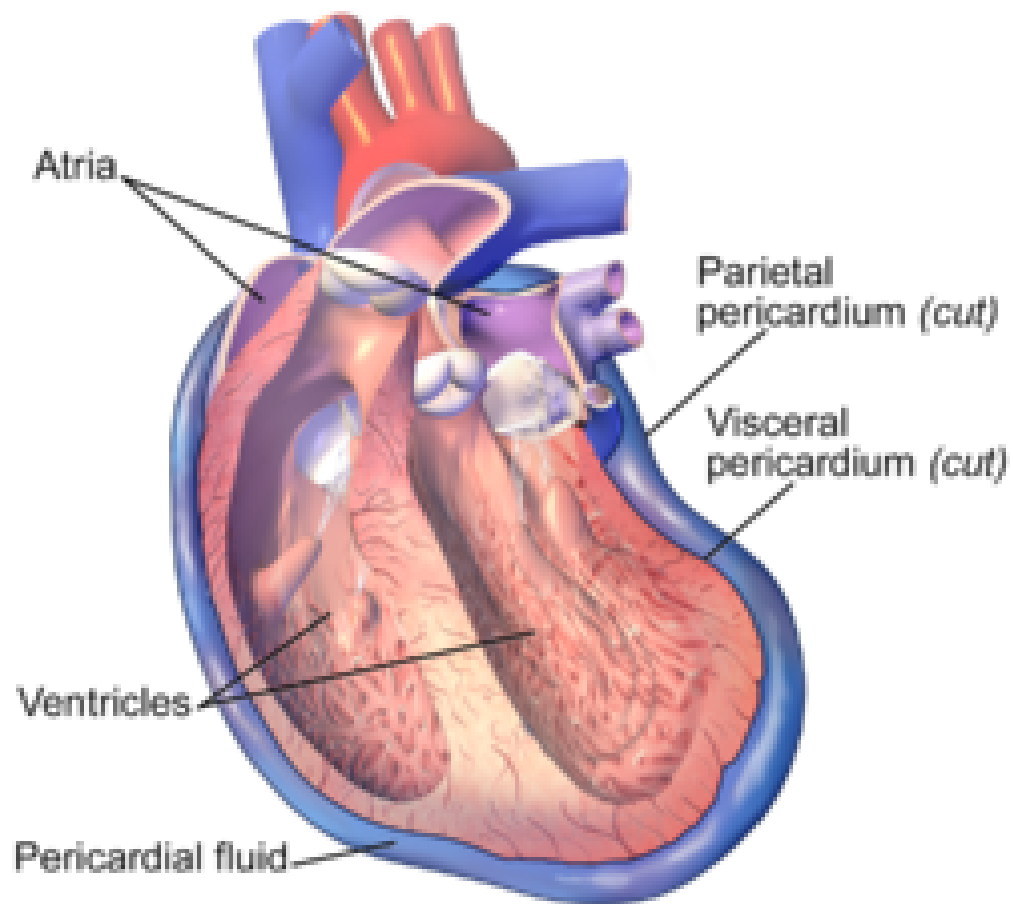
- ⊠ This fibro-serous sac has
- ⊠ 1. Outer fibrous pericardium
- ⊠ 2. Inner serous pericardium.



FIBROUS PERICARDIUM :

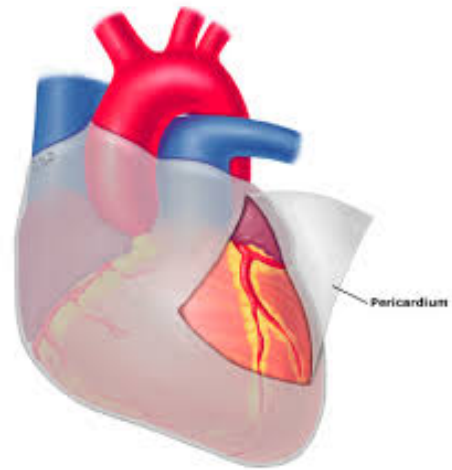
- ⌘ This outer layer of pericardium is a thick, dense layer formed by fibrous tissue.
- ⌘ Anteriorly attached with sternum
- ⌘ Posteriorly related with descending thoracic aorta, esophagus, vagus nerve and sympathetic chain.
- ⌘ On both side related with mediastinal pleura.
- ⌘ Superiorly, enclosing the root of great vessels.
- ⌘ Inferiorly, inseparably attached with the tendinous part of dome of diaphragm





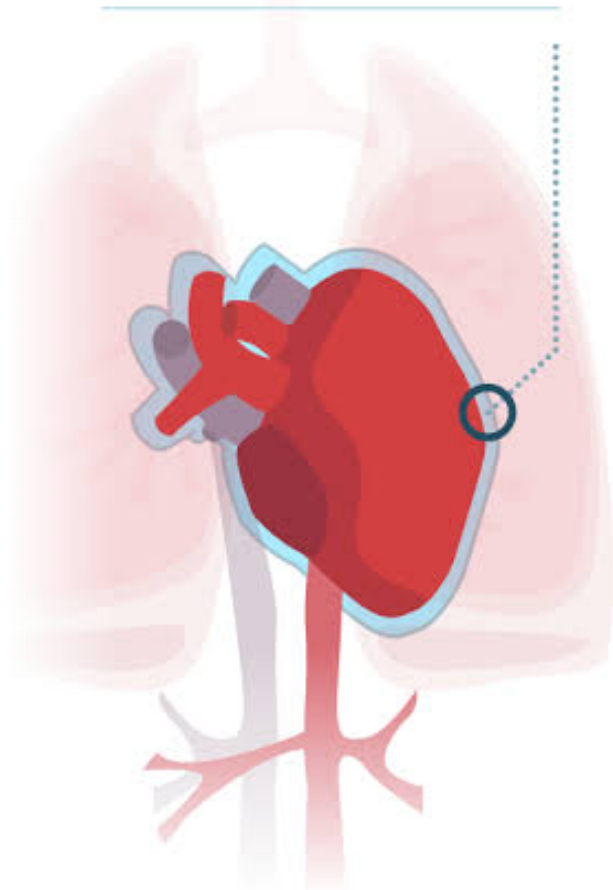
Pericardial Sac



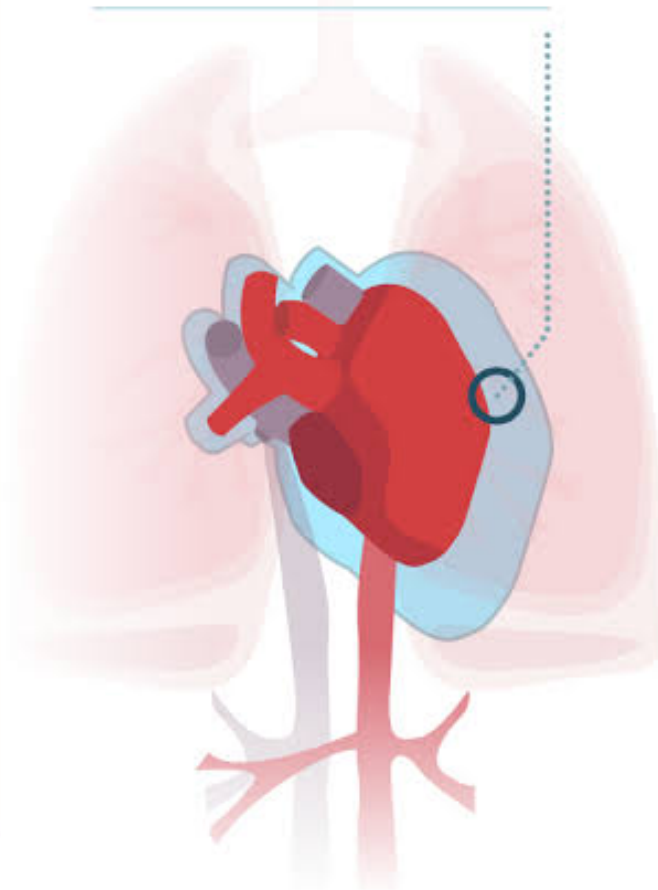


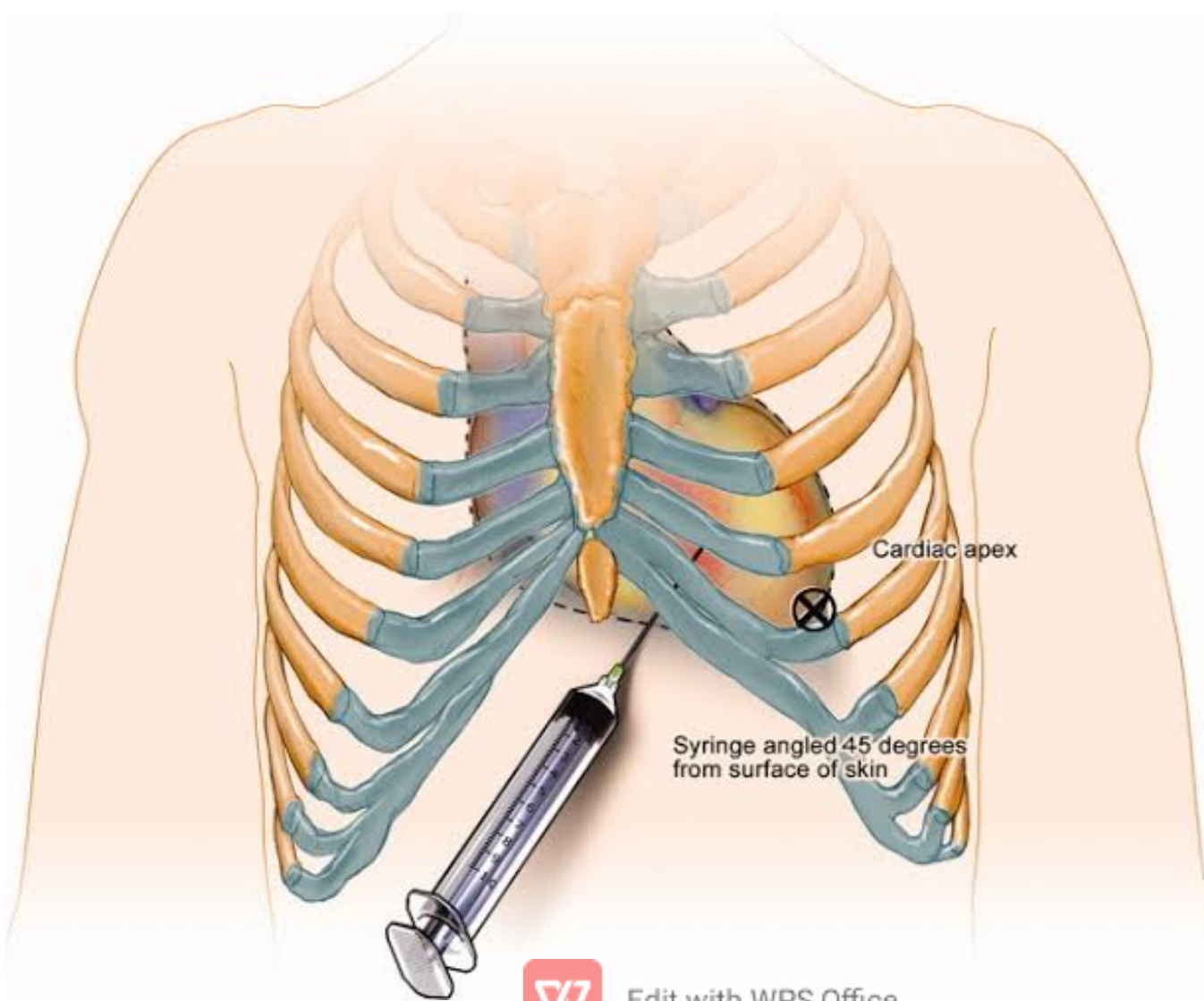
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**Normal Heart
and Pericardium**



**Pericardium With
Pericardial Effusion**

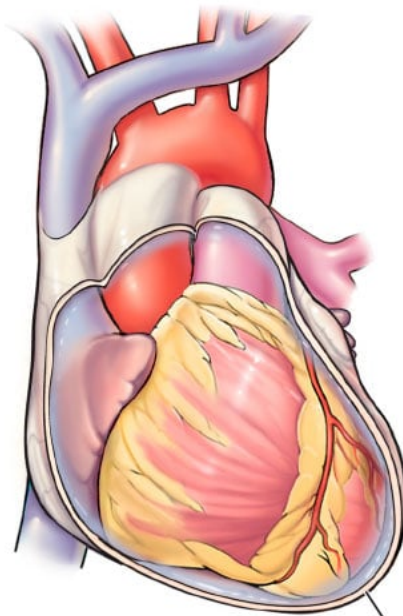




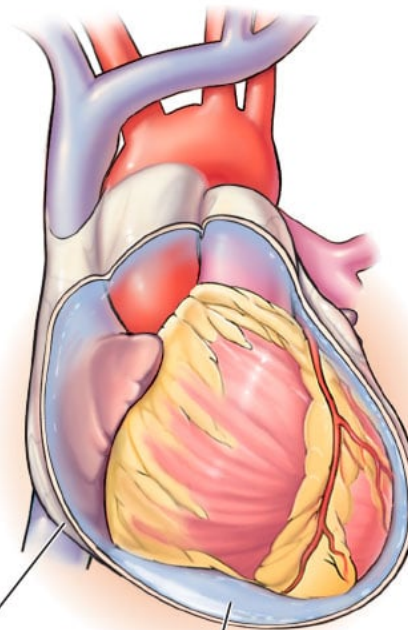
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Pericardial Effusion

Healthy Heart

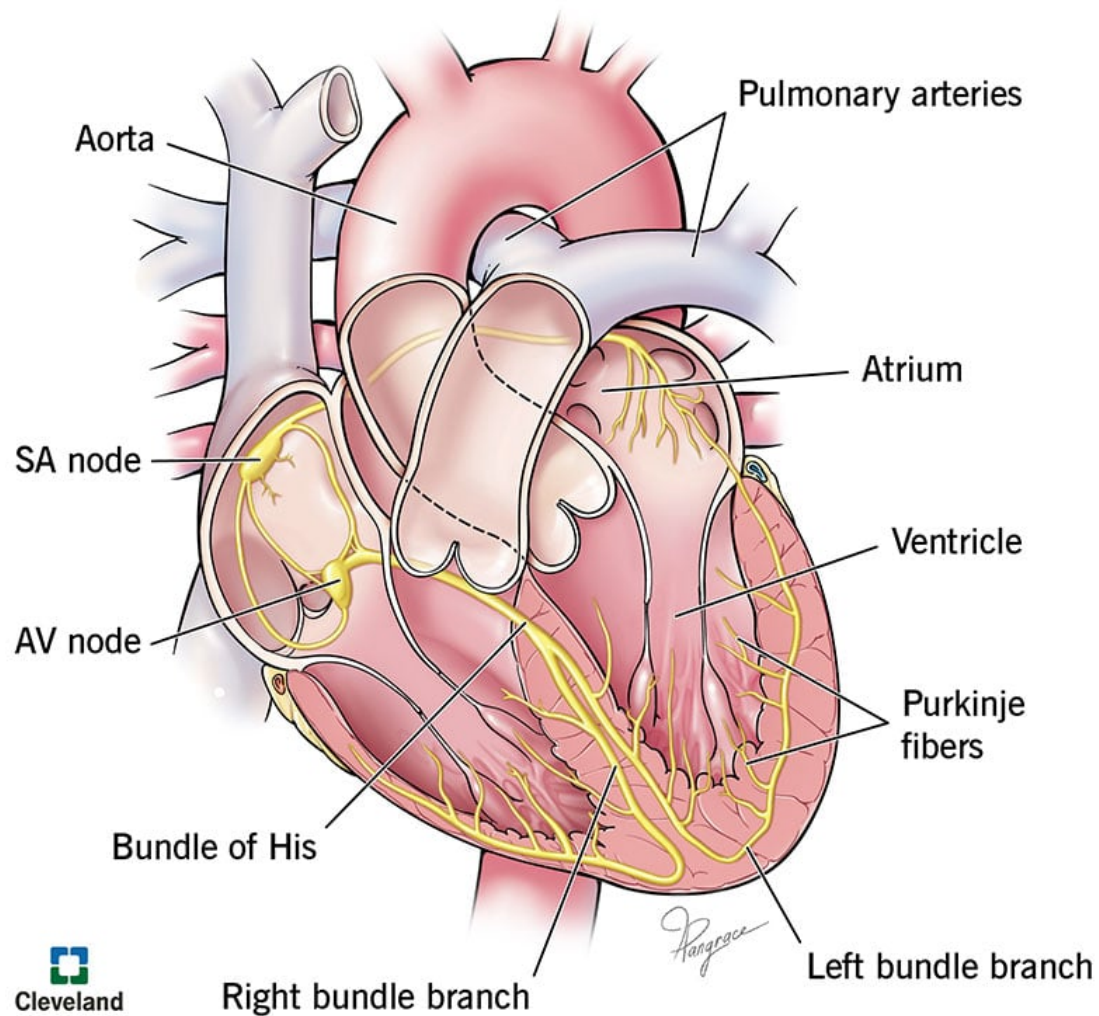


Pericardial Effusion



Pericardium

Excess fluid



SEROUS PERICARDIUM

- ⊠ This inner layer of pericardium again has two layers
- ⊠ 1. Outer parietal pericardium
- ⊠ 2. Inner visceral pericardium.



SINUSES OF PERICARDIUM

- ⌘ Transverse sinus - seen between the arterial and venous limbs.
- ⌘ Oblique sinus - seen between the four pulmonary veins.



BLOOD SUPPLY

- ⊠ parietal pericardium is supplied by pericardio- phrenic artery, intercostal artery and mediastinal artery.
- ⊠ The visceral pericardium is supplied by coronary vessels.



NERVE SUPPLY

- ⊠ parietal pericardium is supplied by somatic nerves ie) 2-6 th intercostal nerves.
- ⊠ The visceral pericardium is supplied by the autonomic nerves,
- ⊠ sympathetic is from T2 to T5 and
- ⊠ parasympathetic from vagus nerve. Hence, it is insensitive to pain.



CONTENTS OF PERICARDIUM

- ✘ 1. Heart
- ✘ 2. Coronary artery and veins
- ✘ 3. Root of great vessels



APPLIED ASPECTS

- ⊠ Pericardial effusion : Collection of fluid in pericardial cavity is called pericardial effusion.
- ⊠ Fluid to be drained by puncturing at the left 5th and 6th intercostal space near the sternum



HEART



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- ⌘ Heart is a muscular pumping organ which pumps the blood and helps for circulation of blood.



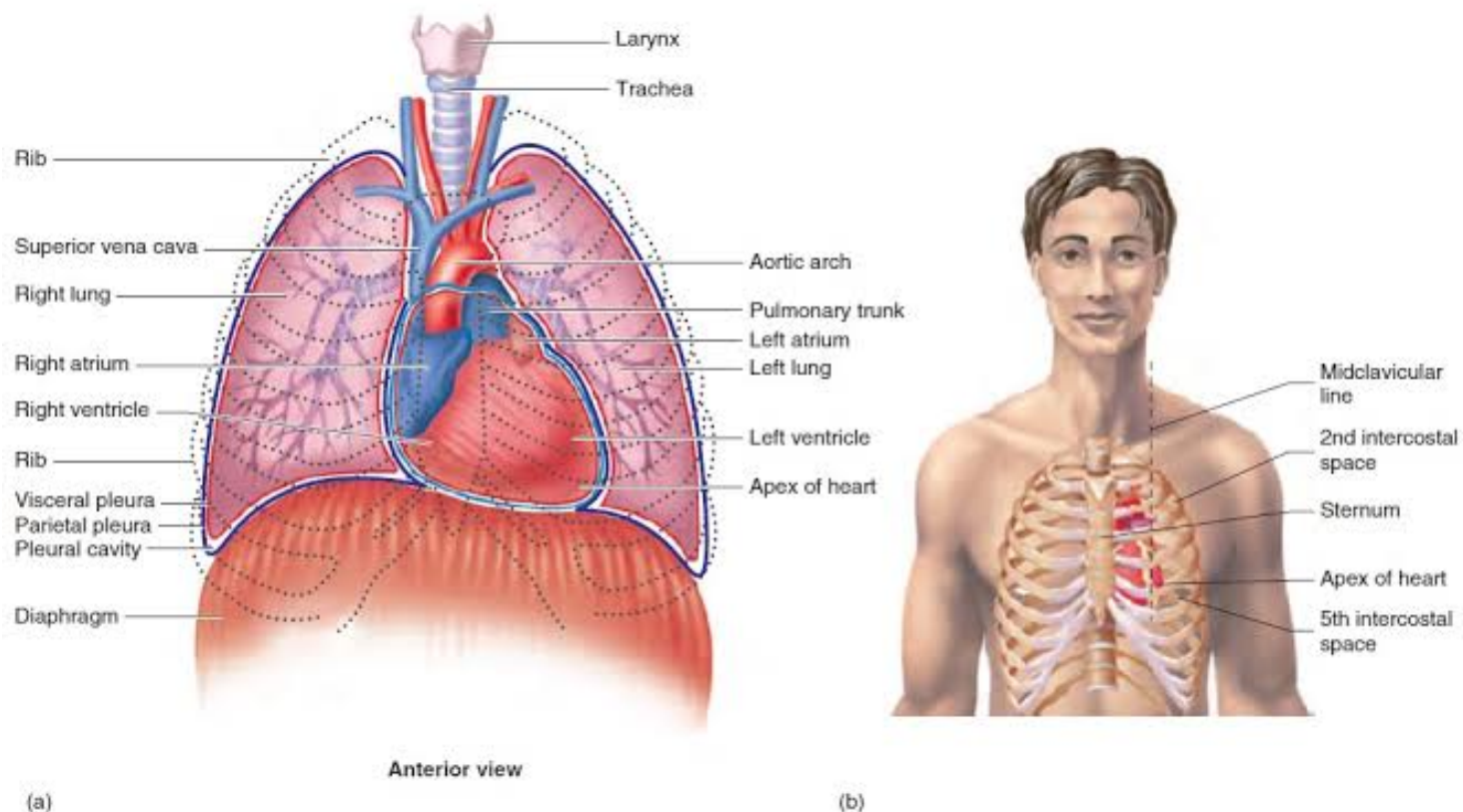
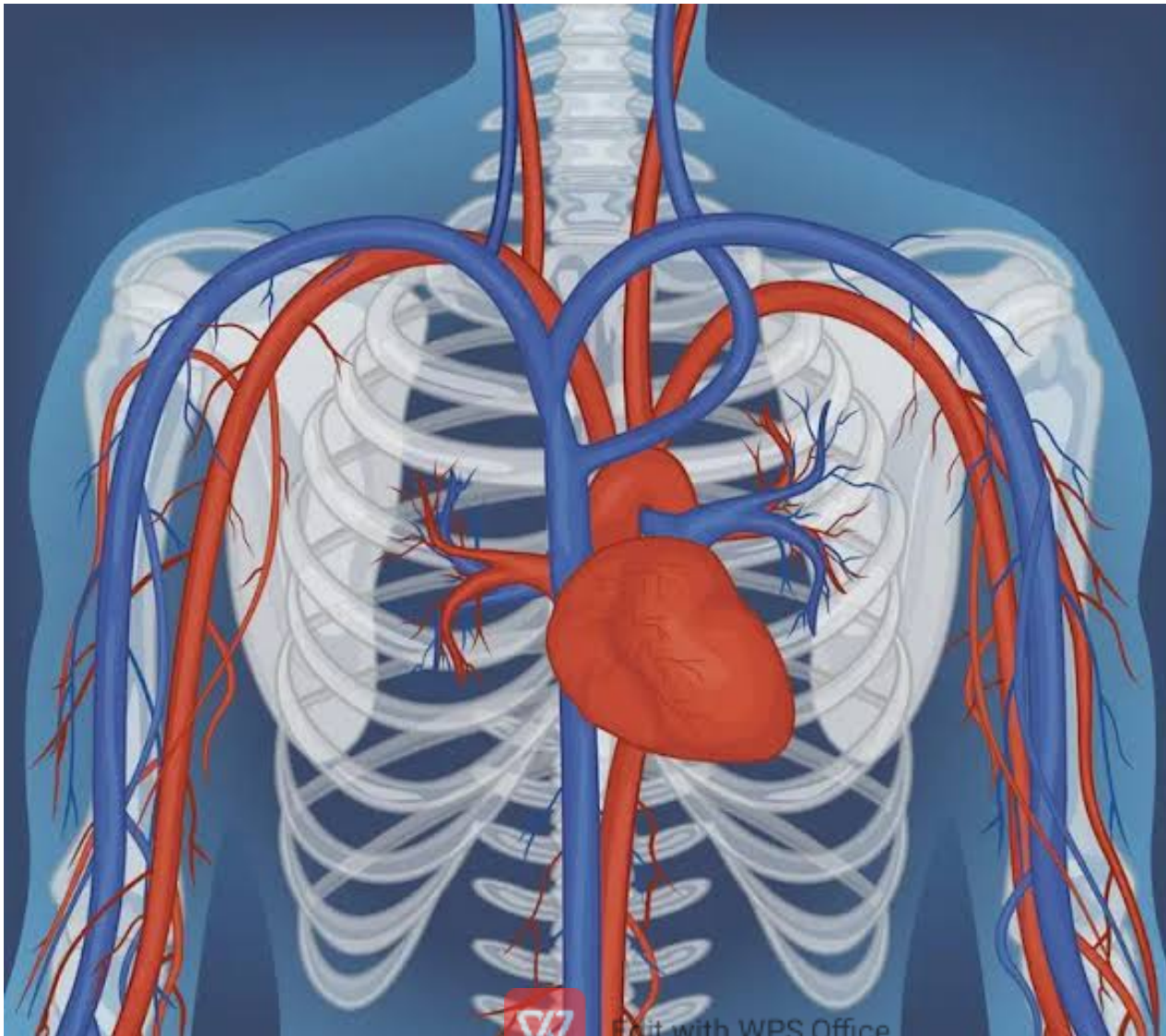


Figure 12.3 **APR** Location of the Heart

(a) The heart is in the thoracic cavity between the lungs, deep and slightly to the left of the sternum. (b) The base of the heart, located deep to the sternum, extends superiorly to the second intercostal space, and the apex of the heart is located deep to the fifth intercostal space, approximately 7-9 cm to the left of the sternum where the midclavicular line intersects with the fifth intercostal space.





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SITE AND SITUATION

- ✕ This situated in the middle mediastinum seated over the dome of diaphragm.



SIZE AND SHAPE

- ⌘ Heart is conical in shape.
- ⌘ And 250gms to 300gms weight and 12 cm in length, 9cm of width and 6 cm of thickness.



EXTERNAL FEATURES

- ⌘ Heart has 4 borders and 4 surfaces

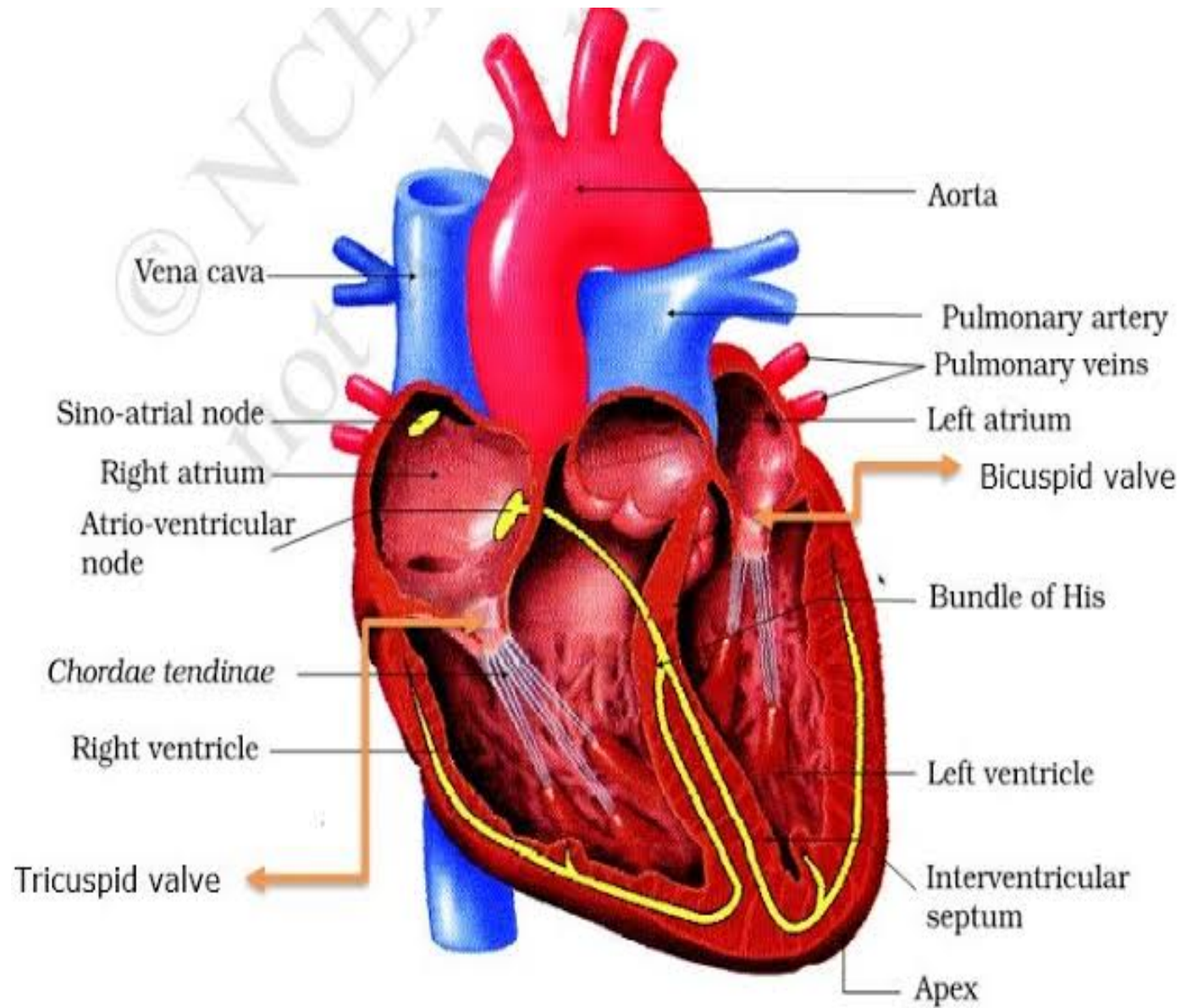
Borders

- ⌘ 1. Right border
- ⌘ 2. Inferior border
- ⌘ 3. Left border
- ⌘ 4. Superior border

Surfaces

- ⌘ 1. Anterior surface
- ⌘ 2. Posterior surface
- ⌘ 3. Left surface
- ⌘ 4. Inferior surface

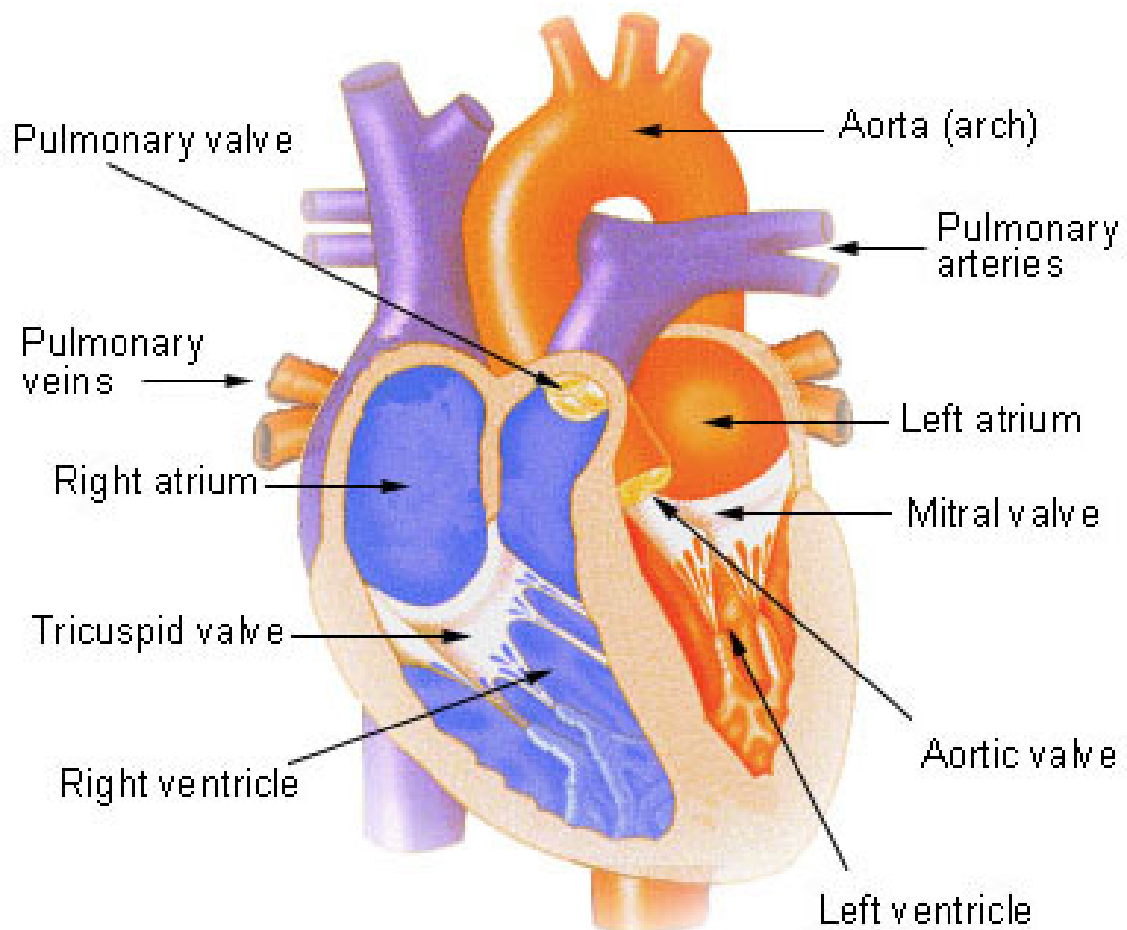


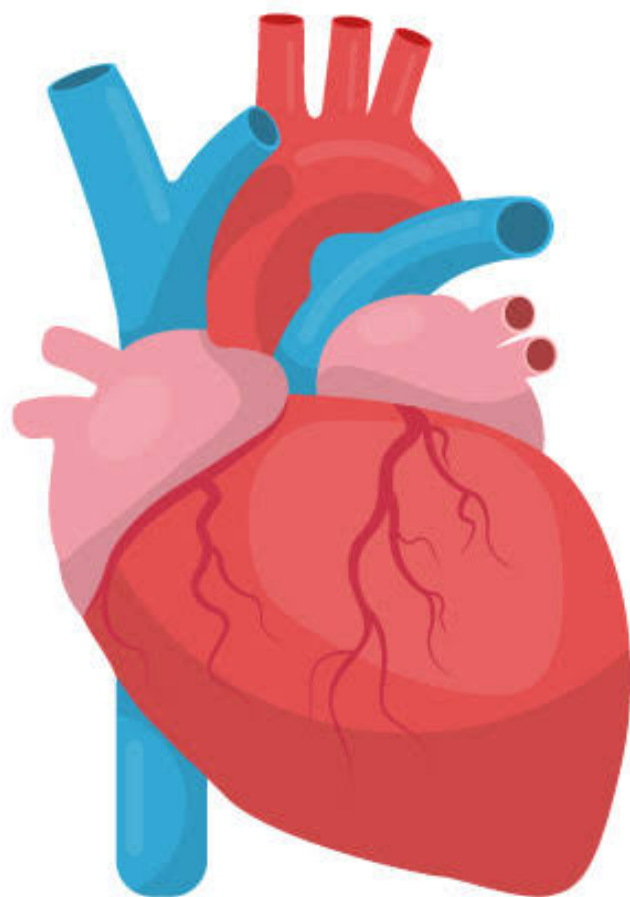


Section of a human heart



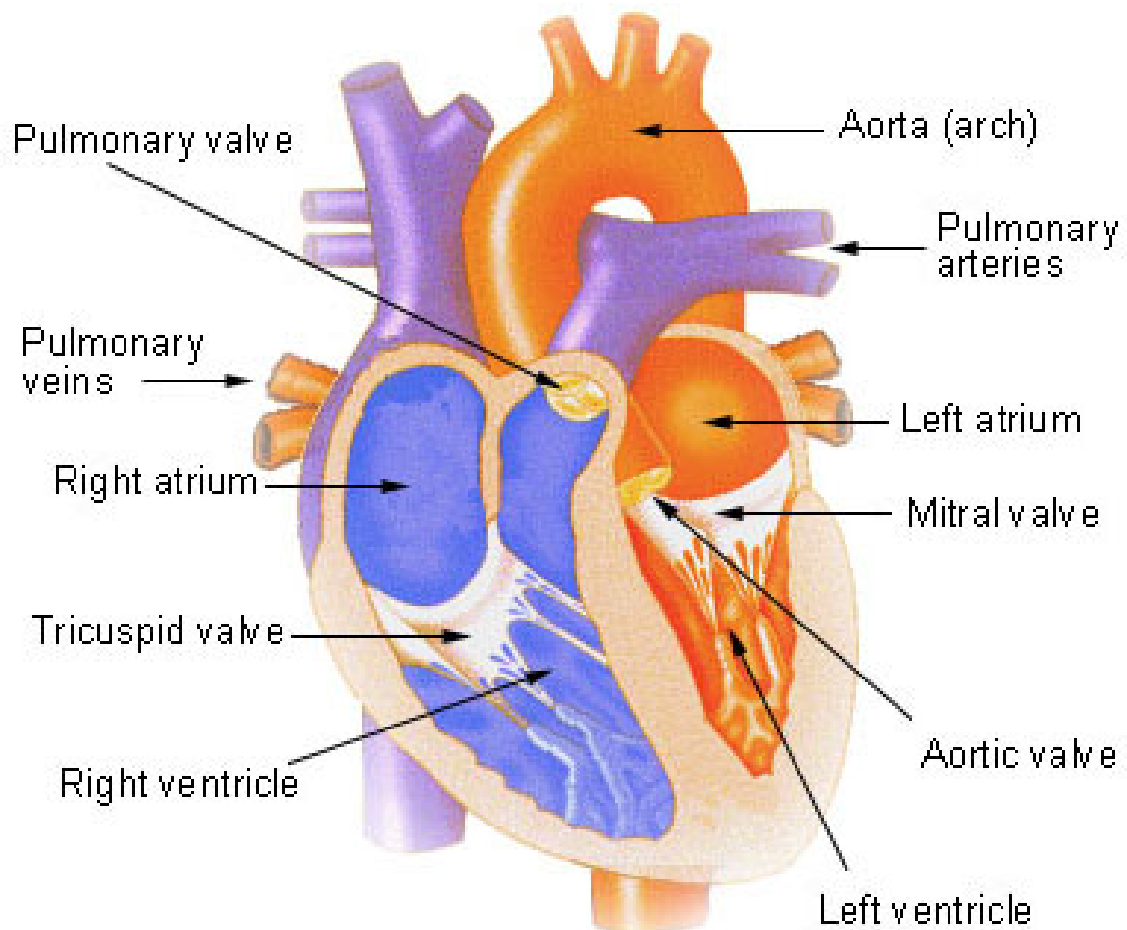
Internal View of the Heart

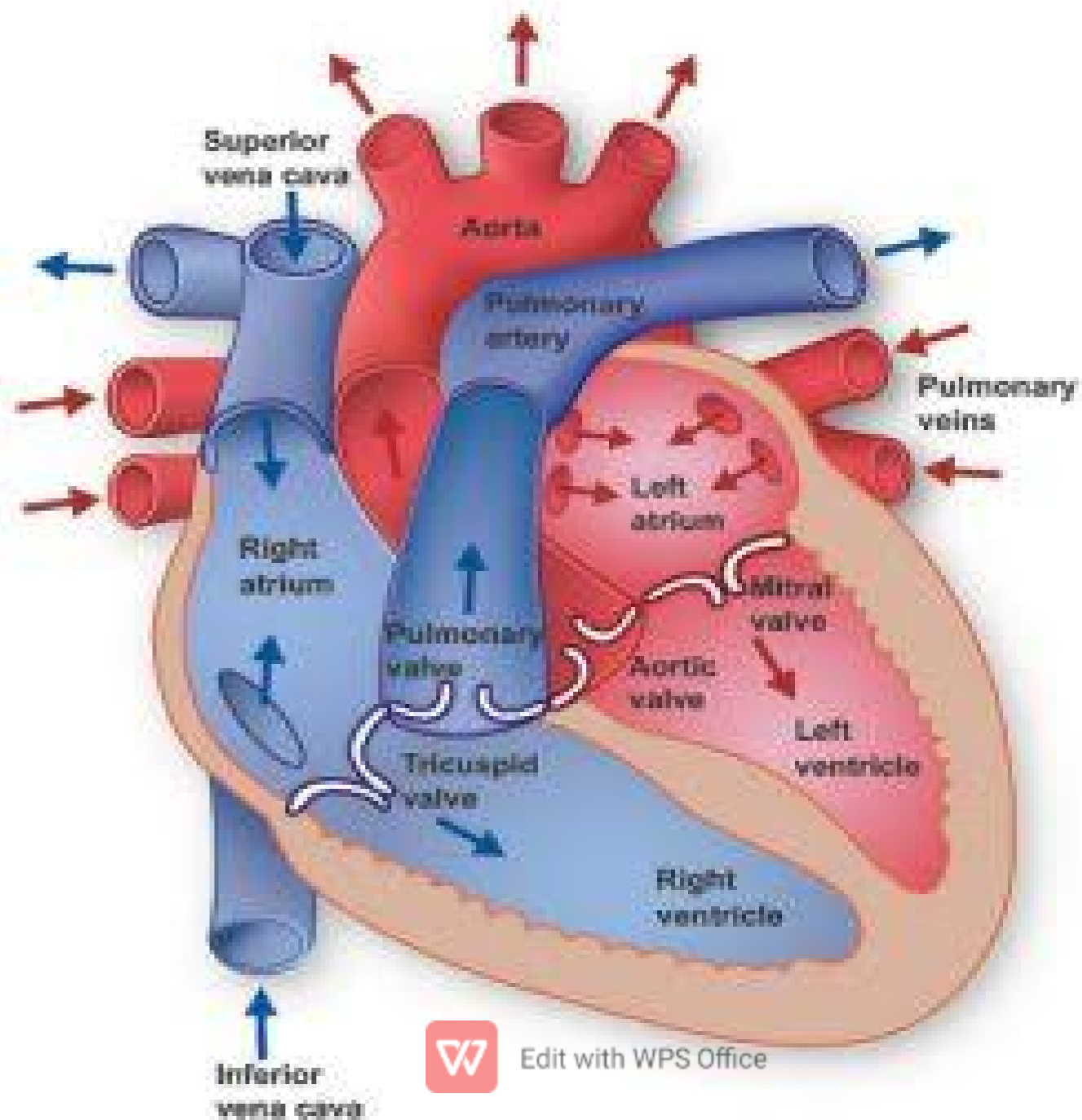


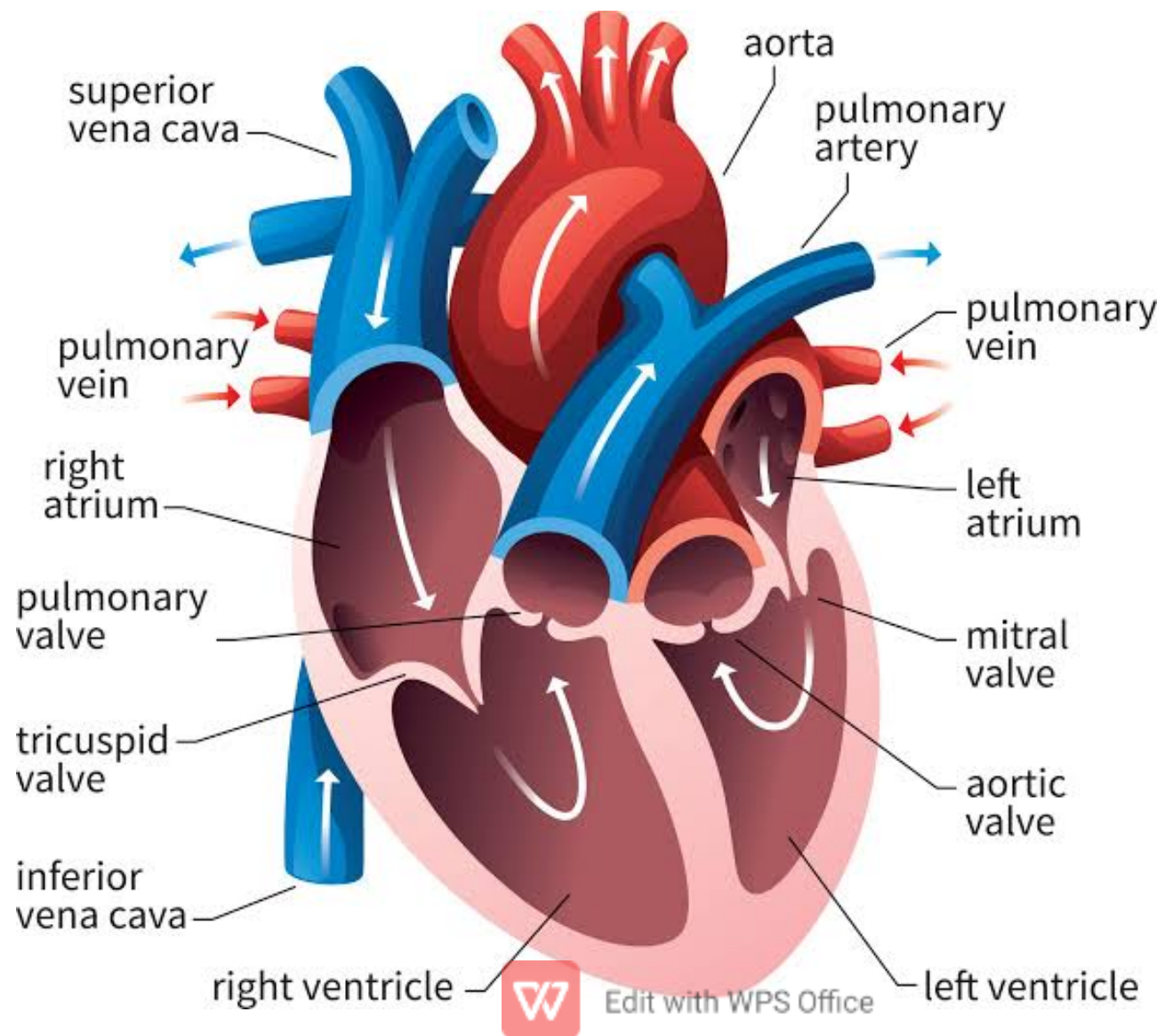


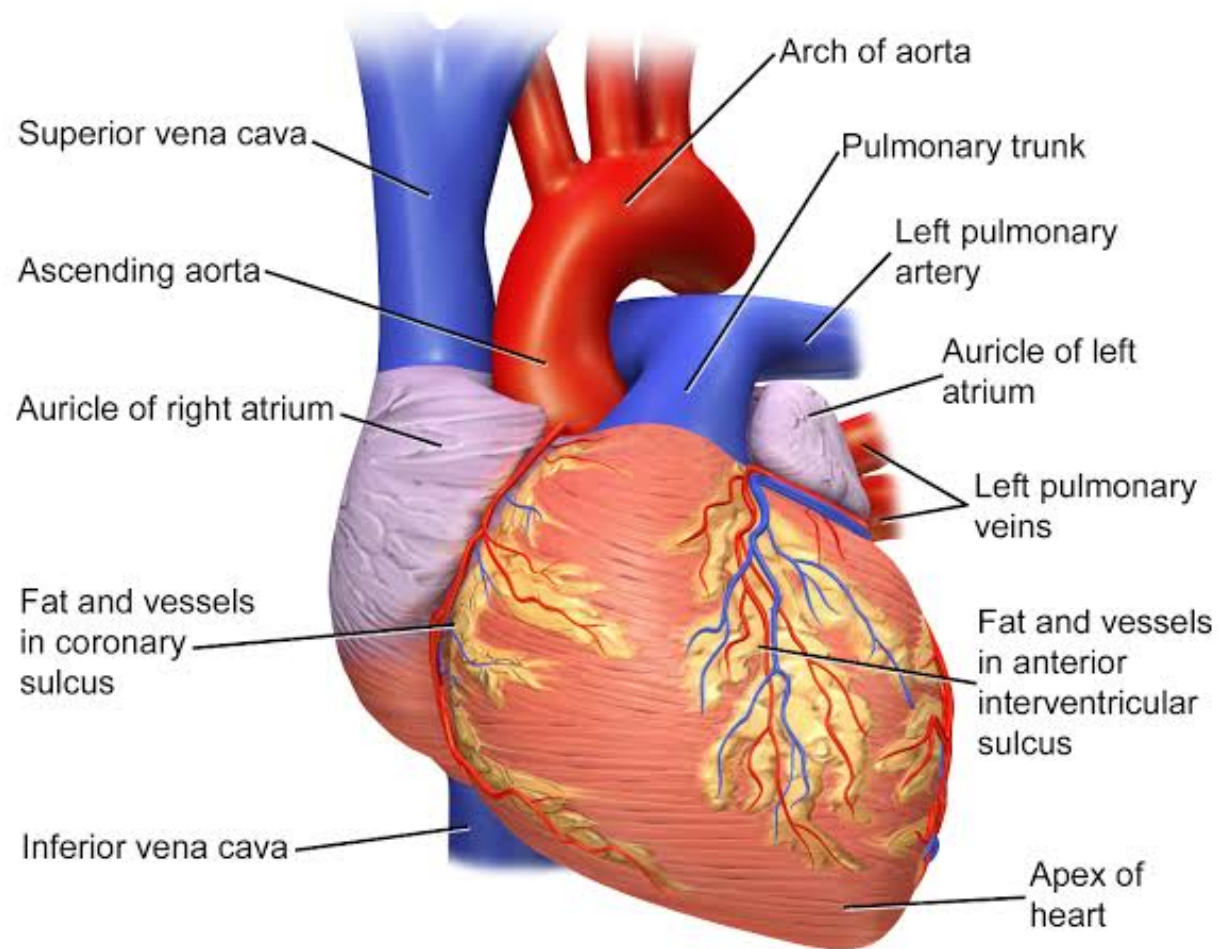
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Internal View of the Heart





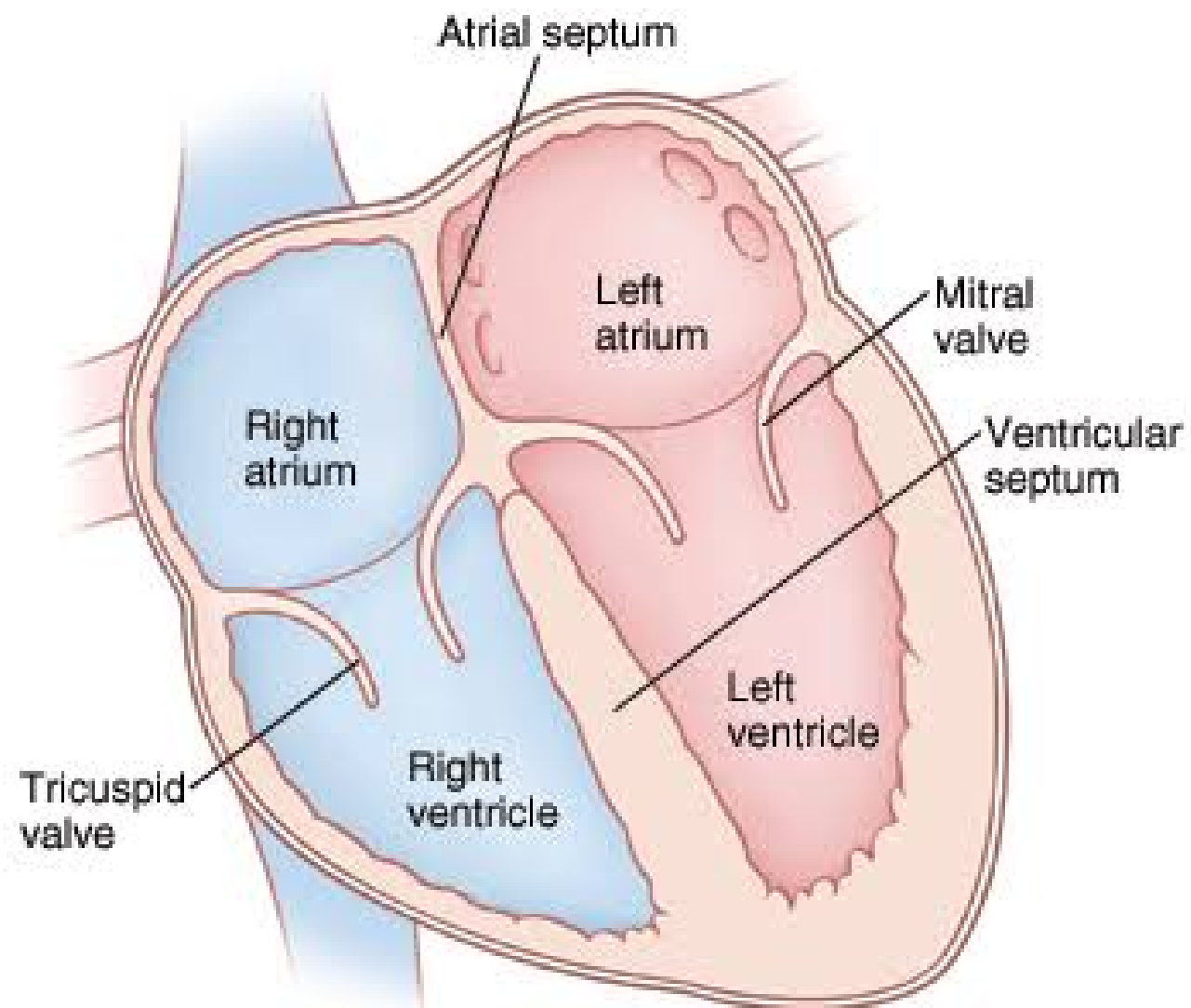


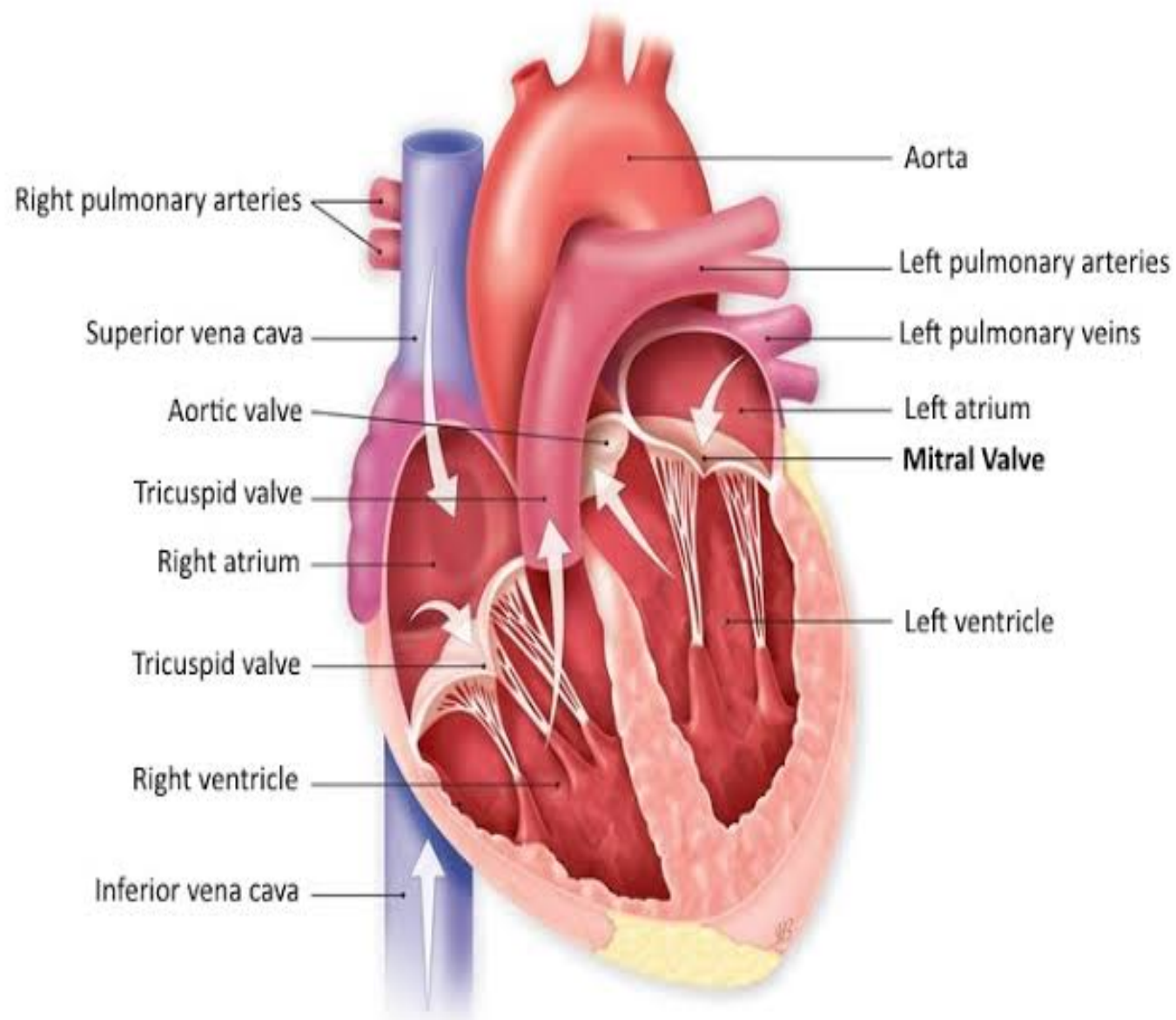


Superficial Heart Anatomy (Anterior)



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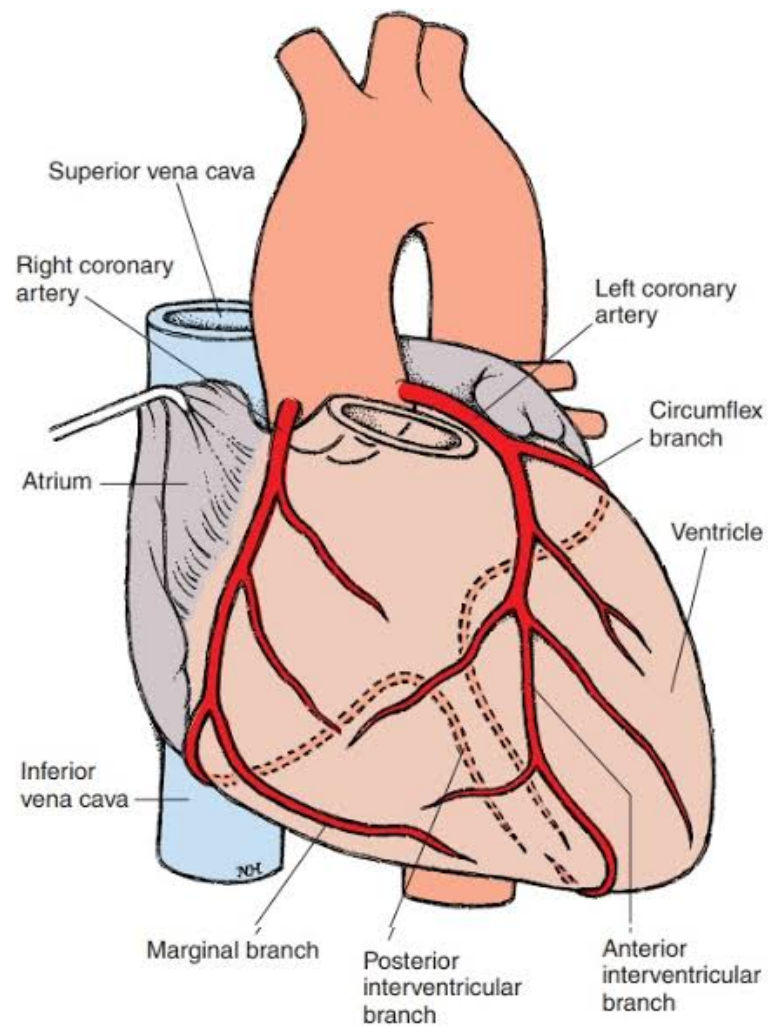


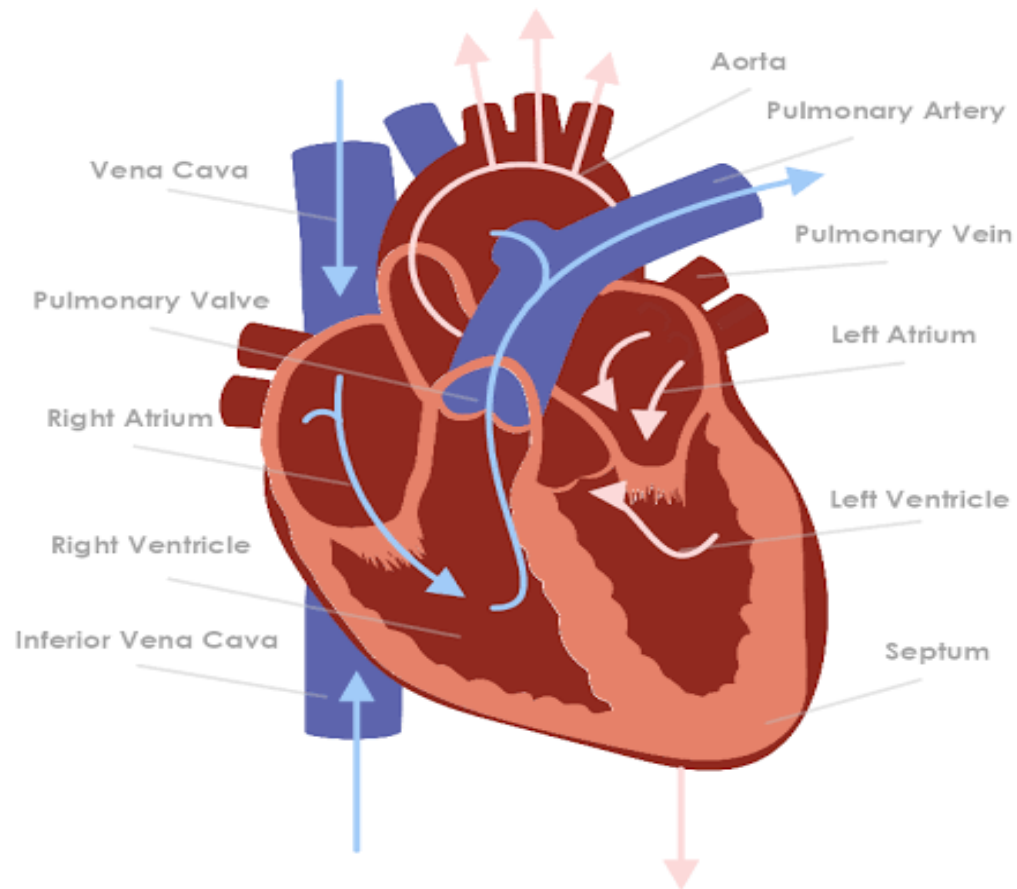
FIGURE
arteries

Coronary Circulation. Anterior view of coronary



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THE HUMAN HEART AND CIRCULATION





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APEX

- ⊠ Heart is conical in shaped and has apex and base.
- ⊠ Apex is seen in the left side.
- ⊠ This is marked a point 9cm left lateral from sternum in 5th intercostal space.
- ⊠ The apex is formed by left ventricle



BASE

✘ Base is seen posteriorly, and it is formed by left atrial chamber.



SULCUS TERMINALIS

- ✘ In right upper chamber, a vertical thickening is seen.
- ✘ This makes a vertical depression in the right margin of heart is called sulcus terminalis.



SULCI AND GROOVE

- ✘ Outer surface of heart has linear depressions called groove or sulci.
- ✘ The two atrial chamber is separated by inter atrial sulci.
- ✘ The ventricles are separated by inter ventricular sulci.
- ✘ The atrioventricular sulcus also known as coronary sulcus which is separated by atrium and ventricle in both side.
- ✘ The interventricular sulcus again divided into anterior and posterior interventricular sulci or groove.



INTERNAL FEATURES

- ⌘ When looking internally, the heart is divided into 4 chambers separated by septa.
- ⌘ both upper and lower chambers are separated by atrioventricular septa.
- ⌘ The upper two atrial chambers separated by inter atrial septum which divide right and left atrium.
- ⌘ Inferiorly two ventricular chambers separated by interventricular septum which divides right and left ventricles.



CHAMBER OF HEART

- ⊠ Heart has 4 chambers namely,
- ⊠ 1. Right atrium
- ⊠ 2. Left atrium
- ⊠ 3. Right ventricle
- ⊠ 4. Left ventricle



RIGHT ATRIUM

- ✘ When looking internally, the right atrium has two areas, they are rough anterior part and smooth posterior part.
- ✘ The rough anterior part is called as pectineal part and smooth posterior part called sinus venorum.
- ✘ Rough part has transverse muscular ridges called musculae pectinae.
- ✘ The transverse ridges begins from a vertical projection is known as cristae terminalis



Smooth posterior part has opening that are superiorly, superior vena cava and inferiorly, inferior vena cava.

And in atrio- ventricular septum, the right atrio - ventricular opening seen which is guarded by tricuspid valve.



LEFT ATRIUM

- ✕ This is seen in posterior part and forming base of heart, mainly mention about four opening which is the opening of the pulmonary veins



RIGHT VENTRICLE

⌘ Right ventricle has rough inflowing part and smooth outflowing part.

⌘ **Rough inflowing part:**

⌘ This is rough due to muscular modification called trabeculae carni. There are three modification namely,

⌘ 1. Ridges

⌘ 2. Bridges

⌘ 3. Pillars



- ✘ Pillars are finger like processes which are seen in 3 groups i.e. anterior, posterior and septal also mentioned as papillary muscles.
- ✘ The apex of the pillars are attached to the cuspid valve by small cord like structure called chordae tendinae.



- ⊠ Smooth outflowing part
- ⊠ This has openings namely,
- ⊠ 1. Right atrio-ventricular opening which is guarded by tricuspid valve.
- ⊠ 2. Opening of the pulmonary trunk.
- ⊠ 3. Anterior cardiac vein, small cardiac vein and thebesian vein



❏ Interventricular septum

❏ This separates the two ventricular chamber. In the right, this has right bundle branch of conducting system and has a septal group of papillary muscles. In the left has the left bundle branch.



LEFT VENTRICLE

- ⊠ As same as right ventricle, it has rough inflowing part and smooth outflowing part.
- ⊠ This has two sets of pillars. Two pillars connecting with two cuspid valves.
- ⊠ The left ventricular wall thickening is three times higher than right ventricle, because the left ventricle pumps blood to systemic circulation i.e. throughout the body.



⊠ Opening

⊠ 1. Left atrio ventricular opening guarded by bicuspid valve.

⊠ 2. Opening of aorta guarded by semi lunar valve.



BLOOD SUPPLY: (CORONARY CIRCULATION)

- ⊠ Heart is supplied by coronary arteries, the major vessels which gives the blood supply to the heart.
- ⊠ There are two coronary arteries namely, right and left coronary arteries.



RIGHT CORONARY ARTERY

- ⊠ Right coronary artery begins from anterior aortic sinus.
- ⊠ Aortic sinus is the small dilatation seen in the anterior part of the root of aorta.
- ⊠ This is the place where the aorta begins.



- ✘ From this passes forwards and downwards towards right between right auricle and pulmonary trunk and passes through the atrio ventricular groove and reaches into right border and inferior part and gives it's branches



- ✘ , 1. marginal artery, further it passes backwards through coronary sulcus and in the meeting point of posterior interventricular sulcus with coronary sulcus, the artery gives another branch,
- ✘ 2. posterior interventricular artery and gets anastomosis with anterior interventricular artery which is the branch of left coronary artery



- ⊠ Right coronary artery gives major blood supply to the heart,
- ⊠ 1. Right atrium
- ⊠ 2. Right ventricle
- ⊠ 3. Posterior 2/3 of the interventricular septum
- ⊠ 4. All the parts of the conducting system of heart except left bundle branch



LEFT CORONARY ARTERIES

- ✕ This begins from left posterior aortic sinus passes between left auricle and aorta and soon divided into anterior interventricular and circumflex artery. The anterior interventricular artery passes through anterior interventricular sulcus and in inferior part gets anastomosis with posterior interventricular artery of right coronary artery of right coronary artery.



- ✧ 1. Nodal branches
- ✧ 2. Ventricular branches
- ✧ 3. Atrial branches
- ✧ 4. Infundibular branches.
- ✧ These arises from various part of coronary artery and gives blood supply to the whole area of the heart.



- ⊠ Left coronary artery supplies,
- ⊠ 1. Left atrium
- ⊠ 2. Left ventricle
- ⊠ 3. Anterior 1/3 of the inter ventricular septum
- ⊠ 4. Left bundle branch of the conductive system of heart.
- ⊠



NERVE SUPPLY OF THE HEART

- ⊠ Heart is supplied by autonomic nervous system.
- ⊠ Sympathetic fibers
 - ⊠ The sympathetic fibers arises from the T2 to T6 of the spinal segments which supplies both lungs and heart the stimulation of sympathetic cause the increase the heart rate which leads tachycardia.
- ⊠ Para sympathetic fibers
 - ⊠ The para sympathetic fibers from the vagus nerve. The parasympathetic reduces the heart rate.



CONDUCTING SYSTEM OF THE HEART

- ⌘ Impulses arising from the higher centers (sympathetic and para sympathetic) unite to form the cardiac plexus and supplies some set of cells which seen over the musculature of the heart which has capacity to generate their own rhythm.
- ⌘ Such, special kind of cells are together called as conductive system of the heart.



CONDUCTING SYSTEM OF THE HEART:

- ⊠ 1. Sino atrial node
- ⊠ 2. Atrio ventricular node
- ⊠ 3. Bundle of his
- ⊠ 4. Right bundle branch
- ⊠ 5. Left bundle branch
- ⊠ 6. Purkinji fibers.



SINO ATRIAL NODE

- ⊠ This is a small mass seen near the atrio venacaval junction i.e. the meeting point of the right atrium and superior vena cava.
- ⊠ This produce 72 beats per minute and heart is functioning in the rhythm of SA node, hence SA node is called as pacemaker of the heart.



ATRIO VENTRICULAR NODE

- ⊠ Impulses from SA node passes through crista terminalis and reaches into atrio ventricular septum, in that have AV node i.e. meeting point of atrium and ventricle, this produces 60 beats per minute.
- ⊠ If heart is functioning in the rhythm of AV node then, AV node is consider as the pace maker of the heart.



BUNDLE OF HIS

- ✘ Bundle from AV node to AV junction mentioned as bundle of His, this passes through the inter ventricular septum and then it divides into right and left bundle branch. Right bundle branch seen in the right side of inter ventricular septum in right ventricular chamber.
- ✘ The left bundle branch seen in left side of inter ventricular septum in left ventricular chamber.



PURKINJI FIBERS

- ✘ From that two branches, fibers reaches to the root of papillary muscles by passes through moderator band, i.e. a connection between the papillary muscles and interventricular septum.
- ✘ From papillary muscles passes through chordae tendinae as purkinji fibers, these fibers reaches into upper part of the cardiac chamber.



VALVES OF THE HEART

- ⊠ Valves are the structure which helps to maintain the unidirectional flow of blood in the heart. if valves are weakened the flow of blood is non unidirectional & cause backflow.
- ⊠ Two types of valves namely,
 - ⊠ 1. Cuspid valves
 - ⊠ 2. Semilunar valves



CUSPID VALVES

- ⌘ This is triangular in shape. The apex is free and attaches to the chordae tendinae, base is attached to fibrous ring surrounding the atrio ventricular opening.
- ⌘ The right atrio-ventricular opening is guarded by tricuspid valve. This has three cusps.
- ⌘ The left atrio ventricular opening is guarded by bicuspid valve and has two cusps.



SEMI LUNAR VALVES

- ⊠ This is a pocket like extension of endocardium. These are the lunar shaped structure which has thickened margin called lunule and central thickening called nodule.
- ⊠ They are guarding the aortic opening, vena caval opening and opening of the coronary sinus.



LAYERS OF THE HEART

- ⊠ The heart has 3 layers namely,
- ⊠ 1. Outer epicardium
- ⊠ 2. Middle myocardium
- ⊠ 3. Inner endocardium



APPLIED ASPECTS

- ⊠ 1. Tachycardia -Increased heart rate
- ⊠ 2. Bradycardia -Decreased heart rate
- ⊠ 3. Arrhythmia -Irregular heart beat
- ⊠ 4. Palpitation -Feeling of throbbing of heart beat



- ✘ 5. Pericarditis -Inflammation of pericardium
- ✘ 6. Myocarditis -Inflammation of myocardium
- ✘ 7. Endocarditis -Inflammation of endocardium



- ✕ 8. failure to pump the blood cause the backwards pressure and leads to congestive cardiac failure.
- ✕ 9. Angina pectoris pain of heart is expressed (referred pain) through pectoral region medial side of arm and shoulder. And this condition is named as angina pectoris. This is the first sign of myocardial infarction.
- ✕ 10. Lesion of myocardium (damage of myocardium) due to lack of blood supply is called myocardial infarction.



- ✘ 11. Any blockage of coronary artery cause the ischemia and form myocardial infarction which commonly known as ischemic heart disease.



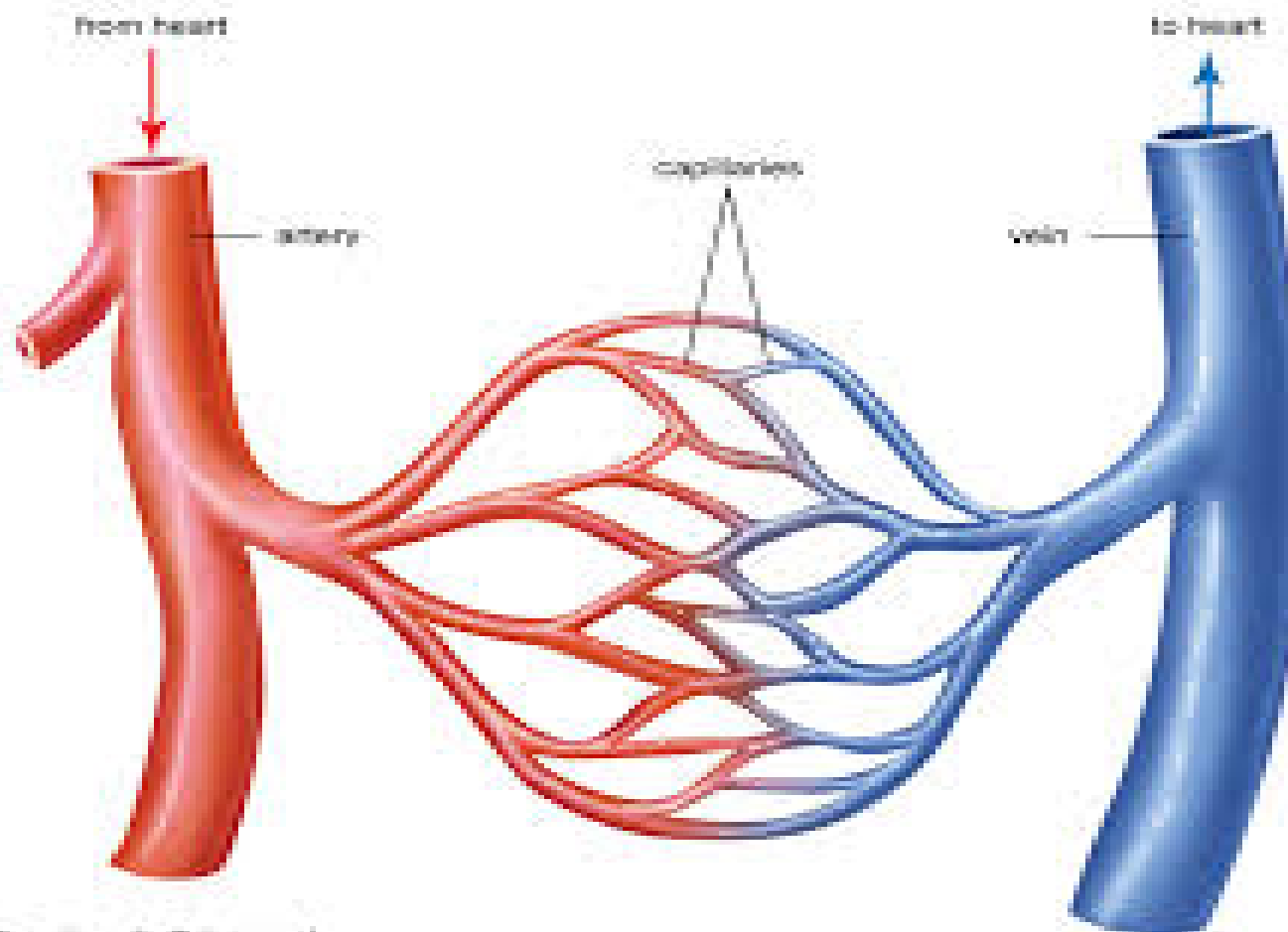
- ✕ 12. Narrowing of valves of heart is called stenosis. Commonly the mitral valve is involved called mitral stenosis. The infective organism beta hemolytic streptococcus cause the mitral stenosis in rheumatic heart disease.
- ✕ 13. Regurgitation valves are the structure helpful to maintain the unidirectional flow of blood. Weakness of the valve will leads back flow of blood. this condition is called as regurgitation.



CIRCULATION



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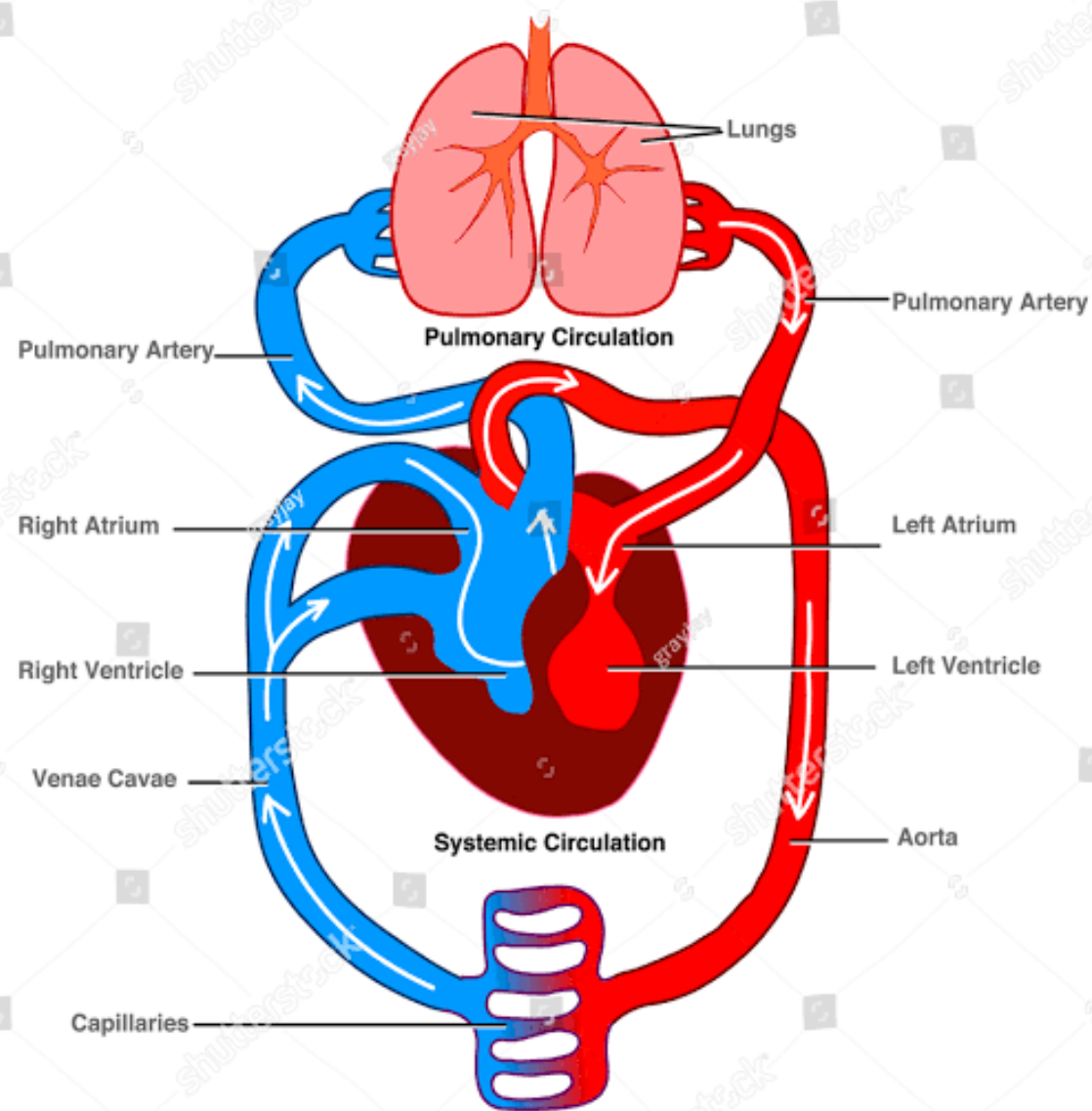


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Heart and Blood Circulation System



⊠ There are commonly three types of circulation seen in the human body and there is a special types of circulation seen in the fetus known as fetal circulation. Here we can see the various circulation of human

⊠ 1. Pulmonary circulation

⊠ 2. Systemic circulation

⊠ 3. Portal circulation

⊠ 4. Fetal circulation



PULMONARY CIRCULATION

- ✘ The pulmonary circulation is the special circulation seen in between the heart and lungs. The pulmonary circulation helps to exchange the gases with blood in the lungs. Here, there is an exception that the pulmonary arteries which carry the deoxygenated blood and pulmonary veins which carry the oxygenated blood.



✘ The pulmonary circulation has the blood vessels

✘ 1. Pulmonary trunk

✘ 2. Pulmonary veins

✘ 3. Pulmonary capillaries

✘ 4. Pulmonary arteries



- ✘ 1. The pulmonary trunk begins from the right ventricle and runs upward and divide into right and left pulmonary artery which carries the deoxygenated blood from heart to the lungs.
- ✘ 2. The four pulmonary veins carries oxygenated blood into heart. After the exchange of gases in respiratory membrane, the pulmonary capillaries are converted in to pulmonary veins which carries the oxygenated blood. From each lung pulmonary veins arises. These pulmonary veins enter into the left atrium.

SYSTEMIC CIRCULATION:

- ✧ The systemic circulation is the circulation in which the blood pumped from left ventricle to entire body surfaces. The systemic circulation is the major circulation which supplies all the parts of the human body. The systemic circulation is studied under two systems namely, arterial system and venous system.



OVERVIEW OF THE SYSTEMIC CIRCULATION:

- ✕ The systemic circulation starts from the left ventricle. While heart pumps the blood, this rushes into the ascending aorta and the ascending aorta the blood enters through arch of aorta and descending aorta, thoracic aorta and then abdominal aorta. The aorta has many arterial branches.



- ✕ These arteries are then converted into the arterioles and the arterioles converted into capillaries which has direct contact with tissues. The capillaries which performs the cellular exchange of the respiratory gases. The capillaries in reverse unites as venules and many venules are connected and forms veins. All the veins are drain into vena cava. There are two vena cava namely, superior and inferior vena cava from the upper part of the body and lower part of the body respectively, This vena cava opens into right atrium.



HEPATO-PORTAL CIRCULATION:

- ⊠ This is a special circulation of the gastro-intestinal tract, which carries venous blood from gastro-intestine tract and spleen to liver.
- ⊠ Portal vein which carries all nutrients from the gastro intestinal tract, and transport to the liver for the metabolism.



FORMATION:

- ⊠ superior mesenteric vein and splenic vein unites and form the hepato-portal vein.
- ⊠ 1. The superior mesenteric vein drain blood from the small intestine
- ⊠ 2. Jejunal, ileal, ileo-colic, right colic, middle colic, pancreatico-duodenal and right gastro epiploic vein drain from the large intestine, stomach and pancreas.
- ⊠ 3. Splenic vein from the stomach, pancreas and part of large intestine.
- ⊠ 4. Right and left gastric vein open directly into the portal vein.



- ✘ This circulation does not connect with heart. All above vein united and form the portal vein from the gastro intestinal tract, gall bladder, pancreas, spleen and transport to liver and open into sinusoids of liver.
- ✘ Then the blood drain into hepatic vein and drain into inferior vena cava.





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AORTA

- ⊠ Aorta is the major vessel carries blood from heart to whole part of the body. This is 2 to 3 cms in diameter. This begins from left ventricle has three parts,
 - ⊠ 1. Ascending aorta
 - ⊠ 2. Arch of aorta
 - ⊠ 3. Descending aorta



- ✘ The ascending aorta arises from the upper part of the left ventricle and it is 5cm in length and covered by the pericardial layer.
- ✘ This runs upwards forwards and towards right then continued as arch of aorta. At the root of the ascending aorta this has three dilatations named as aortic sinuses.
- ✘ They are anterior aortic sinus, right posterior aortic sinus and left posterior aortic sinuses. This gives two coronary arteries namely, right and left coronary artery which supplies the musculature of the heart.



ARCH OF AORTA

- ✕ The arch of aorta is the continuation of the ascending aorta, at the level of T4, right second sternochondral joint, this runs upwards, backwards towards left side and arched over bifurcation of trachea and form the arch. Then it passes downwards at the left side of T4 vertebral level and continued as thoracic aorta.



☒ Branches

- ☒ 1. Brachio cephalic trunk.
- ☒ 2. Left common carotid artery
- ☒ 3. Let subclavian artery.



- ✘ Brachio cephalic trunk

- ✘ This is the first branch of arch of aorta and it runs upwards and immediately divided into right common carotid and right subclavian artery.



- ⊠ Left common carotid artery
- ⊠ This is the second branch of arch of aorta and runs upward, at the level of base of mandible, it divided into the internal carotid artery and external carotid artery.
- ⊠ External carotid artery supplies structures out of the cranial cavity including face and internal carotid artery supplies structures inside the cranial cavity including brain.



- ⊠ Left subclavian artery

- ⊠ This is the third branch of arch of aorta and runs left side in neck and enter into the upper border of the first rib and continuous as axillary artery in left side.

- ⊠ It supplies the root of neck, thyroid gland and muscles of scapular region.



DESCENDING THORACIC AORTA

- ⊠ This is the continuation of the arch of aorta begins from the left side of the body of T4 vertebra and it descends downwards and continued as abdominal aorta at the lower border of the T12 vertebra
- ⊠ Branches of descending thoracic aorta
- ⊠ 1. Nine Posterior intercostal arteries
- ⊠ 2. Bronchial artery
- ⊠ 3. Esophageal artery
- ⊠ 4. Pericardial branch
- ⊠ 5. Superior phrenic artery
- ⊠ 6. Mediastinal branch



ABDOMINAL AORTA

- ✕ The abdominal aorta begins from the T12 vertebral level. This is the continuation of the descending thoracic aorta. This has several branches which supplies the organs and parts of the abdomen and pelvis. This is then divided into Right & Left common iliac artery.



BRANCHES OF THE ABDOMINAL AORTA

- ❑ 1. Inferior phrenic artery
- ❑ 2. Coeliac trunk
- ❑ 3. Middle suprarenal artery
- ❑ 4. Superior mesenteric artery
- ❑ 5. Renal artery
- ❑ 6. Gonadal artery
- ❑ 7. Inferior mesenteric artery
- ❑ 8. Four lumbar artery
- ❑ 9. Median sacral artery

COELIAC TRUNK

- ⊠ This is the artery arises and supplies foregut from the abdominal aorta at L1 vertebral level. It has three terminal branches namely,
 - ⊠ 1. Left gastric artery
 - ⊠ 2. Common hepatic artery
 - ⊠ 3. Splenic artery.



COMMON ILIAC ARTERIES

- ⊠ This is the terminal branch of abdominal aorta and runs downwards and end by external iliac and internal iliac artery at S1 vertebral level.
- ⊠ External iliac artery
- ⊠ This is the terminal branch of common iliac artery and runs downwards and laterally and continuous as femoral artery. It supplies the iliac crest, muscle of pelvis and skin.



INTERNAL ILIAC ARTERY

- ⌘ This is small branch of common iliac artery at sacro- iliac joint, then runs medially, downwards then supplies the pelvic organs, gluteal region and perineum.
- ⌘ Branches
- ⌘ 1. Superior vesical artery 2. Obturator artery:
- ⌘ 3. Middle rectal artery: 4. Inferior vesical artery:
- ⌘ 5. Inferior gluteal artery
- ⌘ 6. Internal pudendal artery
- ⌘ It has the following branches namely,
- ⌘ Vaginal artery- vagina
- ⌘ Uterine artery- supplies the muscle of uterus, vagina and fallopian tube.
- ⌘ Ilio-lumbar artery- supplies the muscle of sacrum
- ⌘ Superior gluteal artery supplies the muscle of gluteal region.



FEMORAL ARTERY

- ⊠ This is the continuation of external iliac artery begins from the level of inguinal ligament then runs downwards and enters into femoral canal, femoral triangle, then enters into adductor canal, then pierces adductor magus muscle and continuous as popliteal artery in popliteal fossa



✘ It gives superficial and deep branches namely,

✘ **Superficial branch**

- ✘ 1. Superficial external pudendal artery
- ✘ 2. Superficial epigastric artery
- ✘ 3. Superficial circumflex iliac artery

✘ **Deep branches**

- ✘ 1. Profunda femoris
- ✘ 2. Deep external pudendal
- ✘ 3. Muscular branches

✘ All these supplies, perineal, thigh region and their muscles.



POPLITEAL ARTERY

- ⊠ This is the continuation of femoral artery seen in the popliteal fossa and divided into two terminal branches. They are,
 - ⊠ 1. Anterior tibial artery
 - ⊠ 2. Posterior tibial artery



ANTERIOR TIBIAL ARTERY

- ✘ This is small terminal branch of popliteal artery and runs downwards and runs along with the muscles of anterior compartment of leg and end as dorsalis pedis artery.



DORSALIS PEDIS ARTERY

- ⊠ This is the continuation of anterior tibial artery and runs along with dorsum of foot and communicated with vessels of sole and form plantar anastomosis. It supplies metatarsal bone and inter tarsal joints.



PERONEAL ARTERY

- ⊠ This is large branch of popliteal artery and gives the muscular branches supplies posterior and lateral compartment of leg.



PLANTAR ARCH

- ⊠ This is the continuation of lateral plantar artery and join with dorsalis pedis artery by its perforating branches gives the digital branches and supplies digits and sole.



ARTERIES OF UPPER LIMB

- ✘ Subclavian artery
- ✘ Subclavian artery is the branch of brachiocephalic trunk in right side and from the arch of aorta in left side.
- ✘ The artery passes laterally and seen behind the sterno clavicular joint and passes lower to the clavicle at the outer margin of first rib. This continuous as axillary artery.



AXILLARY ARTERY

- ⊠ Axillary artery seen in the axillary region, continuation of subclavian artery related with nerves of brachial plexus and axillary vein at axilla at the outer margin of teres minor continuos as brachial artery.
- ⊠ It supplies axilla, pectoral region and breast. The axillary artery divided into three parts by pectoralis minor muscle.



BRACHIAL ARTERY

- ⊠ This begins from the lower part of the teres minor seen in a neuro vascular branch. It runs downwards along the anterior compartment of arm at the cubital fossa divided into two terminal branches namely,
 - ⊠ 1. Radial artery
 - ⊠ 2. Ulnar artery



BRANCHES

- ✕ 1. Profunda brachii
- ✕ 2. Superior ulnar collateral artery
- ✕ 3. Muscular artery : This supplies the anterior and posterior compartment of muscle of arm.
- ✕ 4. Inferior ulnar collateral artery



⊠ Radial artery

- ⊠ This is the terminal branch of brachial artery at cubital fossa. It runs downwards at the lateral side of forearm and then runs in front of the radius in its lower end and continues as deep palmar arch.



ULNAR ARTERY

- ✘ It is the terminal branch of brachial artery and it runs medially downwards in upper part of forearm and then runs vertical in lower part and enter into flexor retinaculum and end by superficial and deep branch.



ARTERIAL SUPPLY OF THORAX

- ✘ Internal thoracic artery
- ✘ It arises from the first part of subclavian artery and runs downwards, forwards medially and terminate in sixth intercostal space by dividing into superior epigastric artery and musculo phrenic artery.
- ✘ It supplies thymus, pericardium, 1- 9 intercostal spaces, breast and rectal sheath.



ARTERIES OF HEAD, NECK AND BRAIN

- ⊠ Common Carotid Artery
- ⊠ The blood supply of the head and neck is by the common carotid artery which is the branch of arch of aorta. There are three major vessels arises from the arch of aorta namely,
 - ⊠ 1. Brachio-cephalic trunk
 - ⊠ 2. Left common carotid artery
 - ⊠ 3. Left subclavian artery.



- ⊠ The brachio-cephalic trunk gives two branches namely,
- ⊠ 1. Right subclavian artery
- ⊠ 2. Right common carotid artery.



- ✕ The common carotid artery seen in the side of the neck region and carrying the blood to the head and neck region. The right common carotid artery arises from the branch of the brachio-cephalic trunk and left common carotid artery is the branch of the arch of aorta.



- ✘ 1. External carotid artery
- ✘ 2. Internal carotid artery.
- ✘ At, the bifurification of the common carotid artery a dilatation is seen called carotid body or carotid sinus.
- ✘ Carotid Body
- ✘ Carotid body has chemorecptors and carotid sinus has the baroreceptors. Both the structure are seen in the dilatation of the common carotid artery.



⊠ The external carotid artery arises from the common carotid artery and enter into the parotid gland and divided into eight branches.

⊠ Branches of the external carotid artery:

⊠ 1. Superior thyroid artery

⊠ 2. Lingual artery

⊠ 3. Facial artery

⊠ 4. Post auricular artery

⊠ 5. Ascending pharyngeal artery

⊠ 6. External occipital artery

⊠ 7. Superior temporal artery

⊠ 8. Maxillary artery.

⊠



INTERNAL CAROTID ARTERY

- ✕ It is the terminal branch of the common carotid artery. It begins from the base of mandible and enters into the cranial cavity through carotid canal. In the cranial cavity, it makes a arterial circle by dividing into the anterior and middle cerebral artery. This arterial circle is called circle of willis.



BASILAR ARTERY

⊠ It is formed by the union of right and left vertebral artery at the lower part of pons. Then it ascends upwards and seen in the basilar sulcus of pons and end by dividing into right and left posterior cerebral artery and it gives

- ⊠ 1. Pontine branch
- ⊠ 2. Labyrinthine branch
- ⊠ 3. Anterior inferior cerebellar
- ⊠ 4. Superior cerebellar
- ⊠ 5. Posterior cerebellar arteries

CIRCLE OF WILLIS

- ⊠ It is a arterial circle seen in the base of brain gives blood supply to brain. It is formed by three sources. They are,
- ⊠ 1. Basilar artery
- ⊠ 2. Right internal carotid artery
- ⊠ 3. Left internal carotid artery.



- ✘ The basilar artery divided into right and left posterior cerebral artery. Internal carotid artery divided into anterior cerebral artery and middle cerebral artery. Anterior communicating artery unites the anterior cerebral artery and posterior communicating artery unites posterior cerebral artery with middle cerebral artery.



VEINS OF HEAD AND NECK

- ⊠ Arrangement

- ⊠ The venous blood from the head and neck are drain in

- ⊠ 1. Internal jugular vein

- ⊠ 2. External jugular vein

- ⊠ 3. Vertebral veins.



VEINS OF BRAIN

- ✕ The venous blood from the brain drains into the dural venous sinus. This dural venous sinus also receives venous blood from scalp and face. The venous blood of the dural venous sinus united and finally drains into the sigmoidal sinus.



SUPERIOR VENA CAVA

- ⌘ Superior vena cava is the largest venous channel carrying venous blood from the upper part of the body to heart. It is formed by the union of the right and left brachio-cephalic vein behind the first sternochondral joint. Then it pierces pericardium at the level of second rib and enter into right atrium at the level of third rib. Before it enters into pericardial sac, it receives azygos vein. It is seen just behind the sternum at the meeting point of ribs also related to sternocostalis muscle and is related with arch of aorta, oesophagus, vagus and phrenic nerve



VEINS OF UPPER LIMB

☒ Cephalic vein

☒ This is the largest superficial vein seen in the upper limb which begins from the lateral limb of the dorsal venous arch seen superficially in the lateral part of the forearm further in front of cubital fossa it communicating with basilic vein by median cubital vein.

☒ Passing through the lateral surface of the arm and seen in the in front of the shoulder joint between the deltoid and pectoralis major has a groove called delto-pectoral groove. So, basilic vein passes through this delto-pectoral groove and opens into the subclavian vein.

✕ Basilic Vein

- ✕ This is the superficial vein seen in the upper limb begins from the medial limb of the dorsal venous arch. It is seen in the medial surface of forearm and makes convexity in the medial aspects in the cubital fossa and then it unites with the cephalic vein by the anti cubital or median cubital vein and further continued as brachial vein.



MEDIAN CUBITAL VEIN

- ✕ This is the communicating vein which communicates the superficial and deep vein seen in front of the cubital fossa. Medially connect with the basilic vein and laterally connect with the cephalic vein at 2.5cm above the lateral epicondyle of humerus. It crosses obliquely and runs superficial, in the cubital fossa. posteriorly related to median nerve, brachial artery, radial artery and ulnar artery.



INFERIOR VENA CAVA

- ✕ It is the largest venous channel carrying venous blood from the lower part of the body to the heart. It is formed by the union of the right and left common iliac vein at the level of L5 vertebra. Then runs upwards and pierces the diaphragm at T8 vertebral level and open into right atrium.



TRIBUTARIES OF INFERIOR VENA CAVA

- ✕ 1. Gonadal vein : in males left testicular vein drains into the left renal vein and then into inferior vena cava. The right testicular vein drain directly into the inferior vena cava. In females the ovarian vein from the left drain into the left renal vein and right ovarian vein drain into the inferior vena cava.
- ✕ 2. Renal vein : Drain blood from the kidney by right and left renal vein respectively,
- ✕ The testicular, ovarian and inferior phrenic vein also drains into renal vein.



- ❑ 3. Supra renal vein : Drain from the supra renal gland
- ❑ 4. Phrenic vein : drain from the diaphragm
- ❑ 5. Hepatic vein: drain from the liver.
- ❑ 6. Lumbar vein: It drains blood from the posterior abdominal wall, vertebra, spinal cord, meninges and both right and left vein drain into inferior vena cava.



LYMPH

- ⊠ Lymph is the special fluid connective tissue. Tissue oozes fluid from the capillary bed is named as lymph .
- ⊠ This is a thin, white colourless fluid. The capillary bed is the area of tissue in which the terminal part of blood vessels seen.
- ⊠ That means, the end of the arterial system has arterial capillaries and beginning of venous system has venous capillaries are seen in the tissues (Capillary bed)

⊠



- ✘ This tissue oozes fluid are collected by some set of vessels which are known as lymphatics or lymphatic vessels.



- ⊠ These lymphatic vessels carries lymph from the tissue bed into lymph nodes. The lymph node are the bean shaped structure has the aggregation of lymphocytes. In the cortex the afferent lymphatic carries lymph from various part of the body. Inside the lymph node the aggregation of lymphocytes named as germinal center is doing the filtration of lymph. That means, the removal of foreign bodies like microbes from the lymph by phagocytosis. The filtered lymph is carried by efferent lymphatic from the lymph node through its hilum. The efferent lymphatics drains into the larger lymphatic vessel or higher lymph node groups.



- ☒ Lymph node groups

- ☒ The lymph nodes are seen in groups.
There are two major lymph node groups,

- ☒ 1. Axillary lymph node

- ☒ 2. Inguinal lymph node



- ✕ The lymphatic vessels drains the lymph into larger lymphatic vessels called lymphatic trunk. These lymphatic trunk drains into the largest lymphatic vessel called thoracic duct.

The lymphatic vessels drains the lymph into larger lymphatic vessels called lymphatic trunk. These lymphatic trunk drains into the largest lymphatic vessel called thoracic duct.



- ✕ The thoracic duct is the largest lymphatic vessel carrying lymph from the lymphatic trunks and drains into the venous system. This has 45 cms of length begins from the cisterna chyli seen in front of the L1 vertebrae. The cisterna chyli is a sac like structure carries lymph from various parts of the body by lymphatic trunk.



- ✕ From the cisterna chyli the thoracic duct ascends upwards and enters in to the thoracic cavity through aortic opening. At the level of T4 vertebra, this crosses from right side of the mediastinum to left and further ascends upwards and forming a hook above the level of clavicle and opens in to the junction between left internal jugular vein and subclavian vein. Throughout its length, this has valves and having a beaded appearance.



✘ The production and maturation of lymphocyte is by lymphoid organs. There are major lymphoid organs they are,

✘ 1. Thymus

✘ 2. Spleen

✘ 3. tonsil



THYMUS:

- ✘ The thymus is the lymphoid gland / lymphoid organ. It is related with production and maturation of the lymphocytes.
- ✘ Structurally, it is bilobed pyramidal shaped structure. In the beginning stage of the life, it is having 5 to 10 gm of weight. After that slowly, the thymus size is increased. In pubertal time, it is 15 to 20 gm of weight. After that it will start retro grade.
- ✘ The growth is reduced and in the middle age of the life (25 to 30), only fibro fatty tissue will present.



☒ Located:

☒ The thymus is located in the anterior mediastinum behind the manubrium sterni in the thorax especially in the lower part of the neck.



☒ Relation

☒ Anterior : manubrium sterni

☒ Posterior : trachea, oesophagus, arch of aorta, phrenic and vagus nerve and its branch recurrent laryngeal nerve are seen.



- ⊠ Blood supply
- ⊠ The thymus is supplied by the internal thoracic artery.
- ⊠ Nerve supply
- ⊠ The thymus is controlled by the stellate ganglion.



❏ Function

- ❏ The thymus is for production and maturation of the WBC in childhood. After that, the other lymphoid organs spleen, bone marrow will functioning and produced the WBC.



SPLEEN

- ✘ The spleen is the organ of lymphatics and has linked to blood vascular system. It plays important role in production, storage and destruction of the red blood cells.



- ✘ Site and situation

- ✘ It is a wedge shaped organ situated in the left hypochondrium and epigastric region. It is present between the fundus of stomach and diaphragm



✘ Size and shape

- ✘ It is 12.5 in length, 7.5cm in width and 2.5cm in thickness and has purple in colour. It is soft in consistency and present obliquely in the ribs. (Interestingly it has 1 inch of thickness, 3 inches of width, 5 inches of length, 7 ounces of weight and situated between the 9 to 11 th ribs)



☒ External features : This has the two ends, three borders, two surface and two angles and hilum.

☒ Ends

☒ 1. Anterior end : This is directed forwards and downwards present in mid axillary line.

☒ 2. Posterior end : This is rounded and directed upwards and backwards, present over the upper pole of right kidney.



✕ Borders

- ✕ 1. Superior border: This has a notch
- ✕ 2. Inferior border: it is rounded and blunt.
- ✕ 3. Intermediate border: it is rounded in nature, seen in the visceral surface.



☒ Surface

- ☒ 1. Diaphragmatic surface: it is convex and smooth
- ☒ 2. Visceral surface: it is concave and rough in nature.

☒ Angle

- ☒ 1. Antero-basal angle: it is present between the superior border and anterior end.
- ☒ 2. Postero - basal angle: it is present in between the inferior border and posterior end.



- ✘ Hilum: it is present between the superior and intermediated border. The splenic artery, splenic vein and tail of pancreas seen this region.



RELATION

- ⊠ Peritoneal relation
- ⊠ 1. Gastro-splenic ligament : It attach the hilum and greater omentum of stomach.
- ⊠ 2. Lino-renal ligament : It attach the hilum and anterior surface of the right kidney.
- ⊠ 3. Phrenic ligament : It support the anterior end of the spleen.



☒ Visceral relation

- ☒ 1. Fundus of stomach
- ☒ 2. Anterior surface of the right kidney
- ☒ 3. Splenic flexure of colon
- ☒ 4. Tail of pancreas
- ☒ 5. Diaphragm



⊠ Arterial supply: It is supplied by the splenic artery which is the branch of the coeliac trunk.

⊠

⊠ Venous drainage: From the hilum of the spleen and drain into the portal vein

⊠ Lymphahtic drainage : Drains into the pancreatico-splenic lymph node.

⊠ Nerve supply : Sympathetic from the coeliac plexus.



- ⊠ Function
- ⊠ 1. Phagocytic action
- ⊠ Spleen produces the monocytes and has the action of phagocytosis which engulf the pathogenic and foreign organism.
- ⊠ 2. Hemopoiesis
- ⊠ Spleen is the production unit of red blood cells even during the intra uterine life.
- ⊠ 3. Storage of red blood cells
- ⊠ The spleen stores the red cells and behaves as blood bank.
- ⊠ 4. Destruction of red blood cells
- ⊠ The spleen act as a graveyard of RBC which destruct the cells while entering into the minute capillaries of spleen.
- ⊠ 5. Immunity: It stimulate and produces the antibodies of relevant antigen.



DISORDERS OF IMMUNE SYSTEM

- ✧ 1. Allergic reaction: it is a immunological reaction occur in body due to any allergic agents causes the itching, rashes develop in skin.
- ✧ 2. Auto immune disease: it is due to loss or failure to recognize the body own tissue and foreign tissue cause the self destruction of the own tissue. Example- myasthenia gravis, systemic lupus erythematosus.



- ✘ 3. Acquired immune deficiency syndrome : it occurs due to HIV virus. It is commonly occur in homosexuality, mother to foetus recurrent blood transfusion and unsterile intravenous injection. It is serious & fatal diseases .





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